CSE4203: Computer Graphics

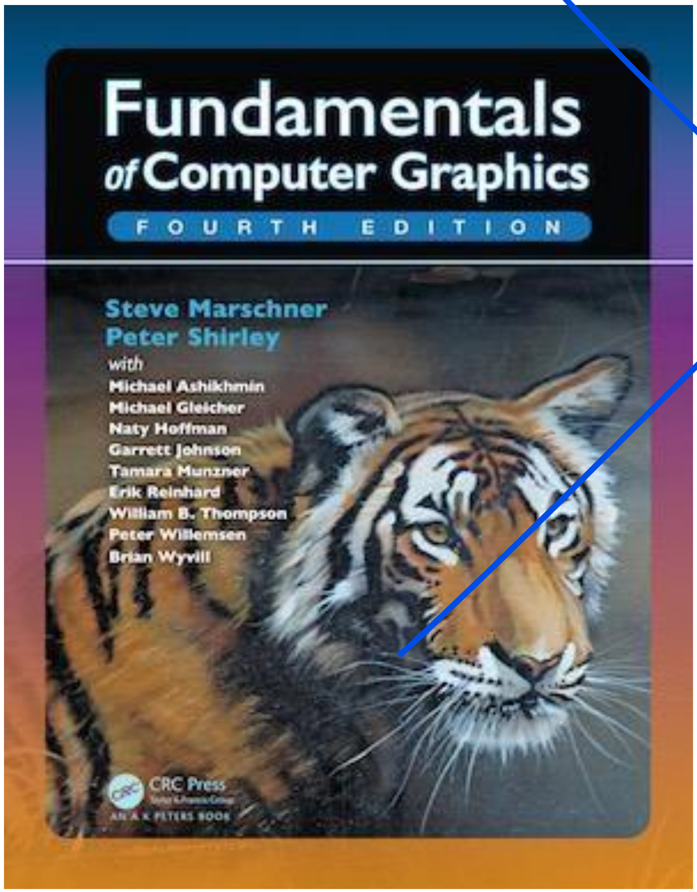
Chapter – 6 (part - A)

Transformation Matrices

Outline

– Transformation
– Linear Transformation 1
  • Scaling
  • Shearing
  • Rotation
– Composite Transformation 2

Credit



CS4620: Introduction to Computer Graphics

Cornell University

Instructor: Steve Marschner

http://www.cs.cornell.edu/courses/cs4620/2019fa/

2D Linear Transformations (1/1)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

Linear Transformation: Operation of taking a **vector** and produces **another vector** by a **simple matrix multiplication**.

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (1/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

- S_x and S_y is called the scaling factors which determines how much scaling is applied along the x and y axis

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (2/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

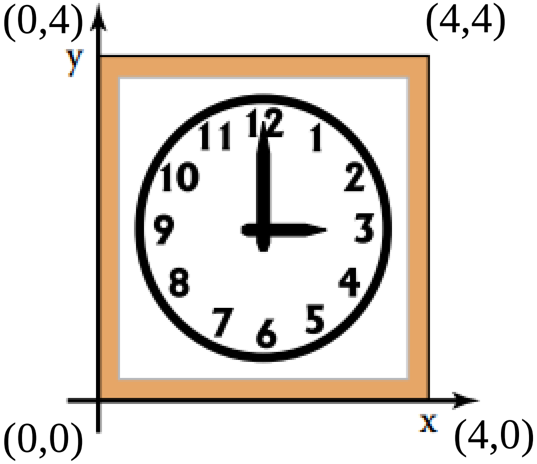
The matrix does to a vector with Cartesian components (x, y):

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (3/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



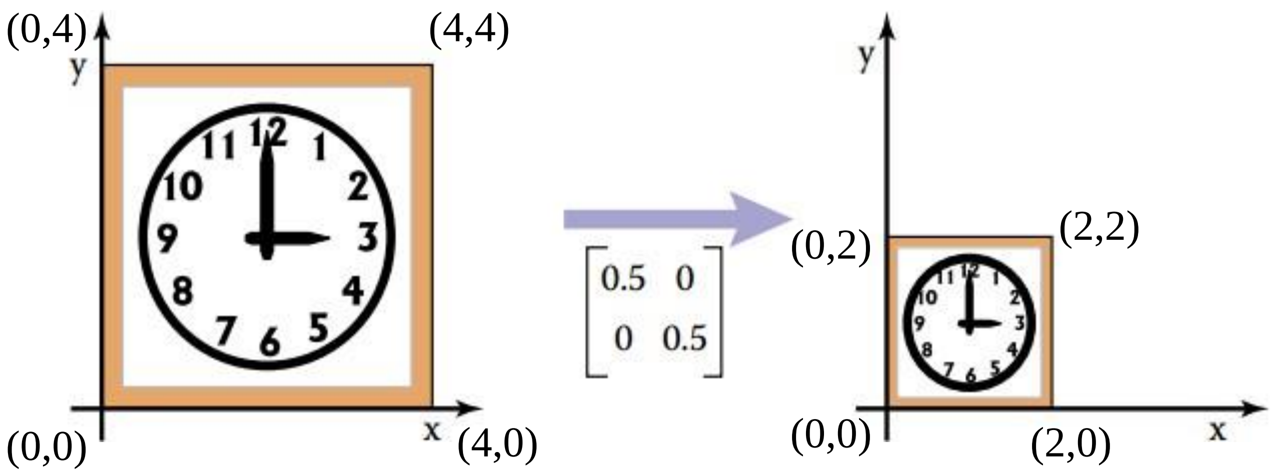
$$\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (4/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 4 & 0 & 4 \\ 4 & 4 & 6 & 0 \end{bmatrix}$$

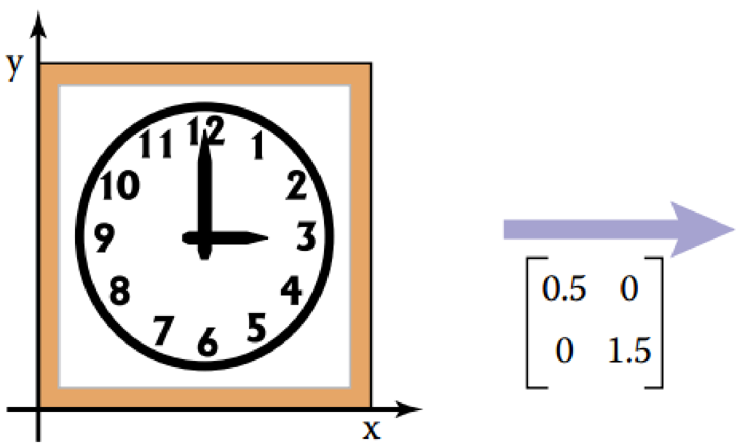
$$= \begin{bmatrix} 0 & 2 & 0 & 2 \\ 2 & 2 & 6 & 6 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (5/6)

$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



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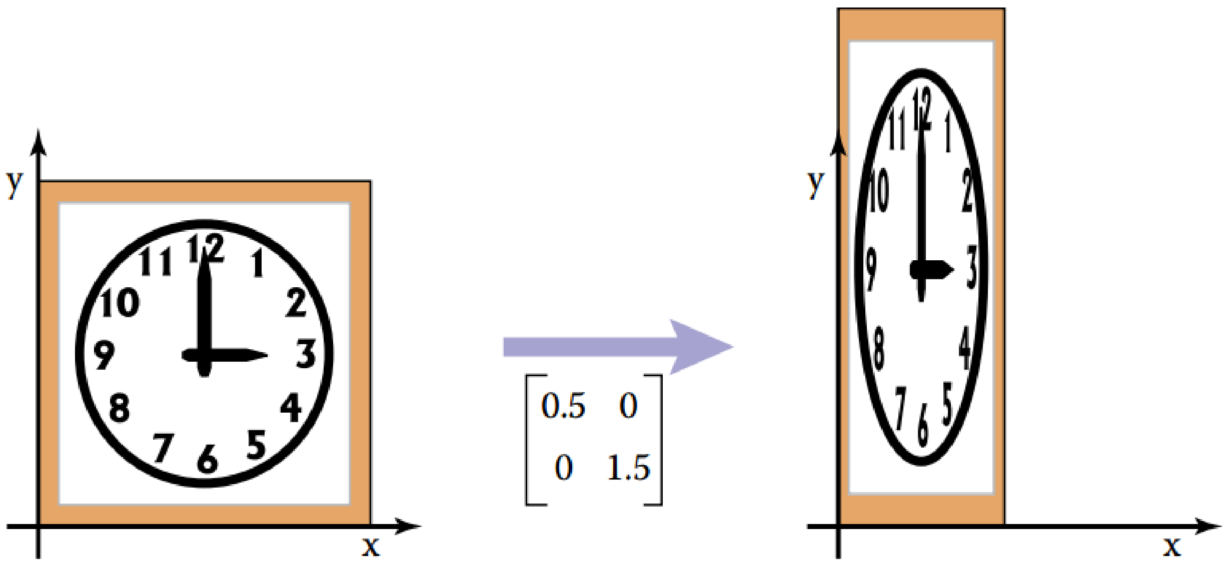
Scaling (6/6)

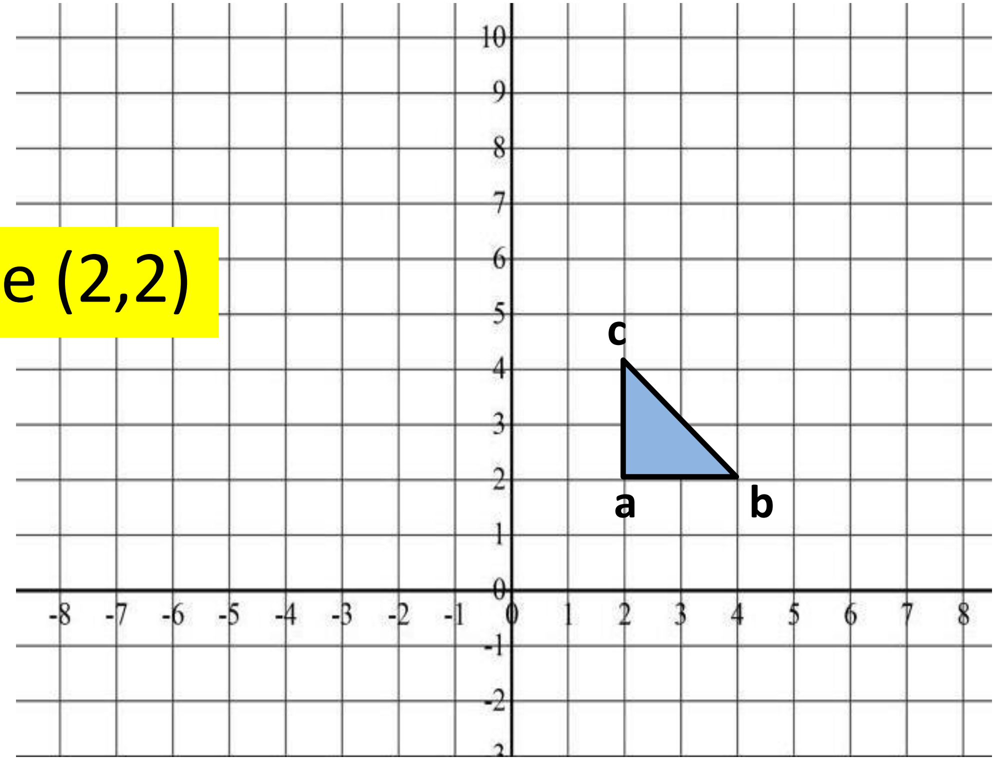
$$\mathrm{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

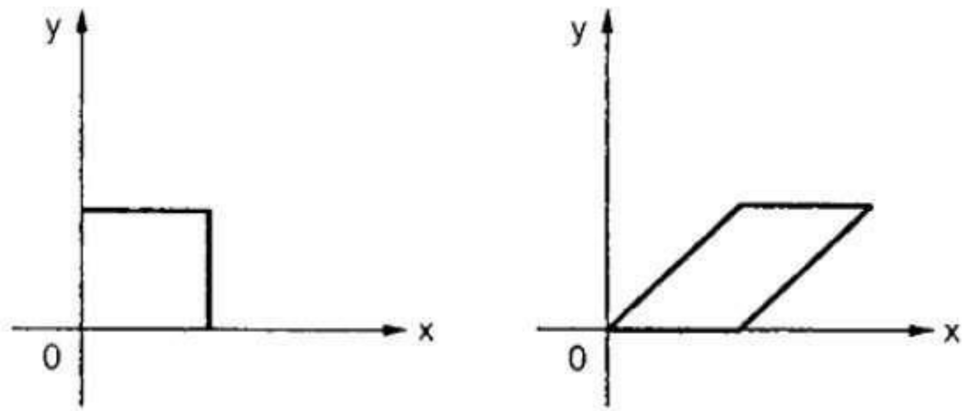
Scaling (6/6)

Scale (2,2)



Shearing (1/5)

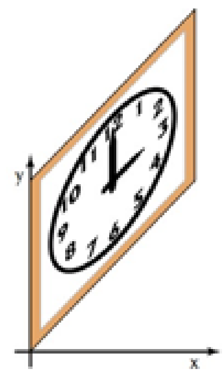
- A shear is something that pushes things sideways



(a) Original object (b) Object after x shear

Shearing (2/5)

- A shear is something that pushes things sideways

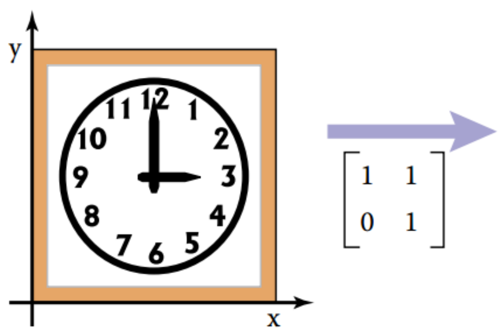


$$\text{shear-x}(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \quad \text{shear-y}(s) = \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix}$$

Source: https://www.tutorialspoint.com/computer_graphics/2d_transformation.htm

Shearing (3/5)

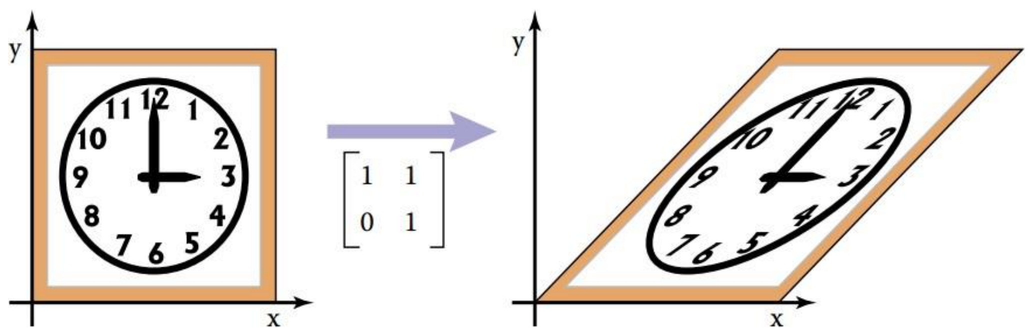
$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Shearing (4/5)

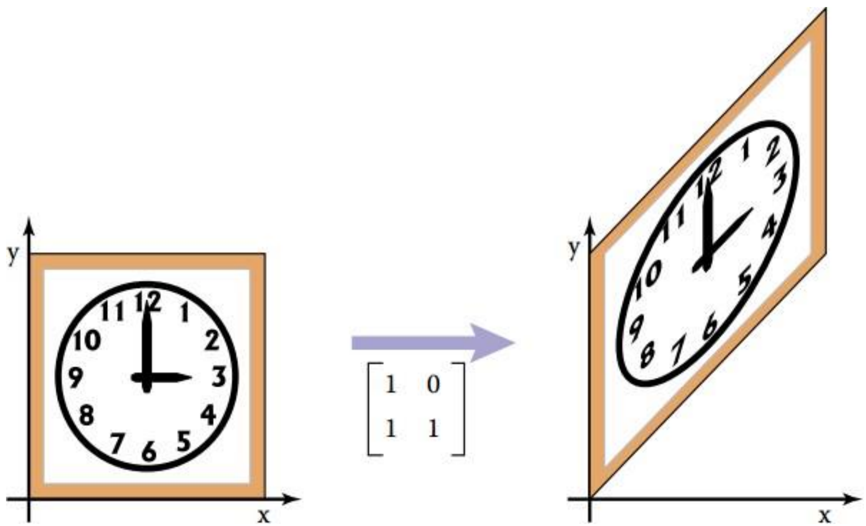
$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Shearing (5/5)

$$\text{shear-y}(1) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

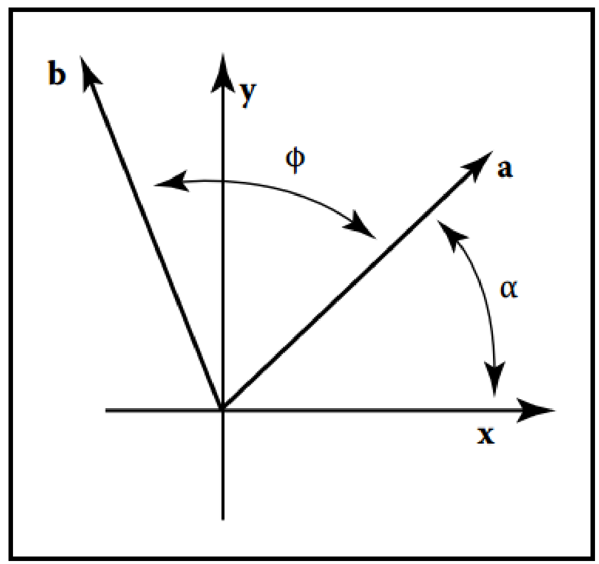


Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (1/11)

$$x_a = r \cos \alpha,$$

$$y_a =$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (2/11)

$$x_a = r\cos\alpha,$$

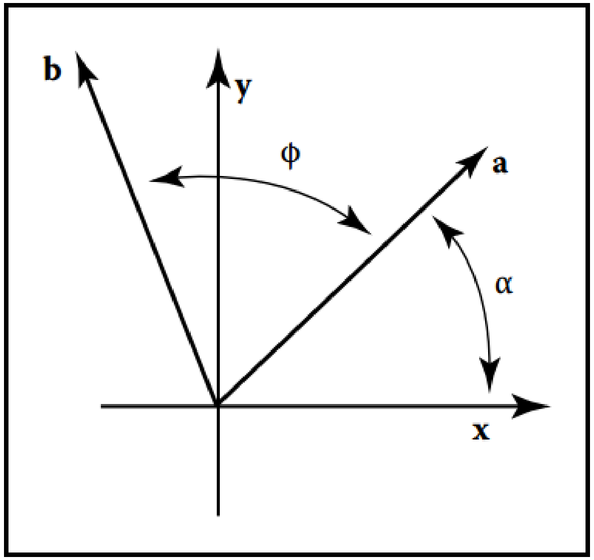
$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha + \phi) =$$

$$y_b =$$

Anti-clockwise rotation = (+) rotation

Clockwise rotation = (-) rotation



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

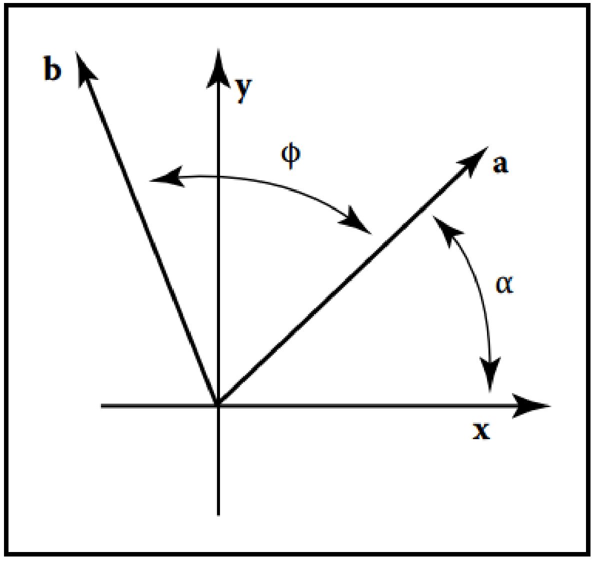
Rotation (3/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha + \phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b =$$



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Rotation (4/11)

$$x_a = r\cos\alpha,$$

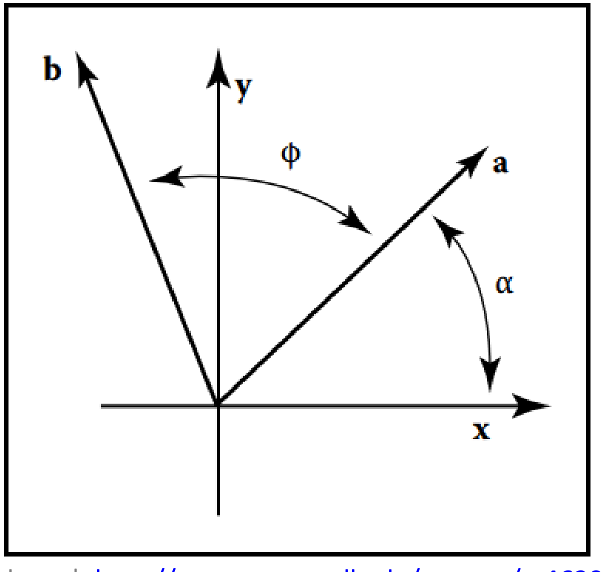
$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b =$$

$$y_b =$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (5/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

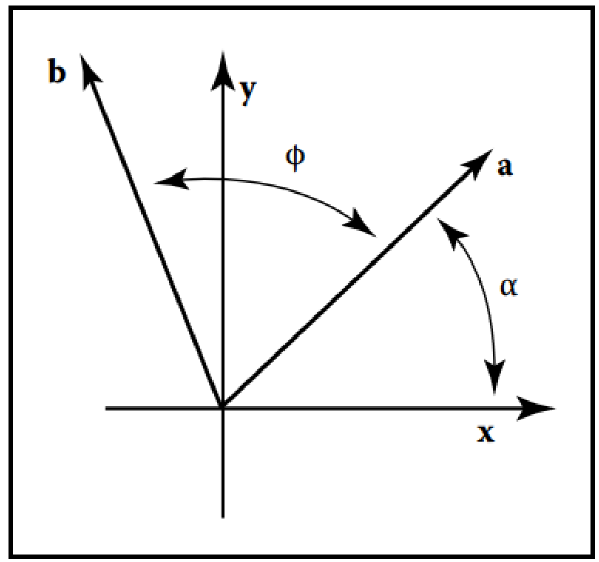
$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b = x_a\cos\phi - y_a\sin\phi,$$

$$y_b = y_a\cos\phi + x_a\sin\phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} & \\ & \end{bmatrix}$$



Rotation (6/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

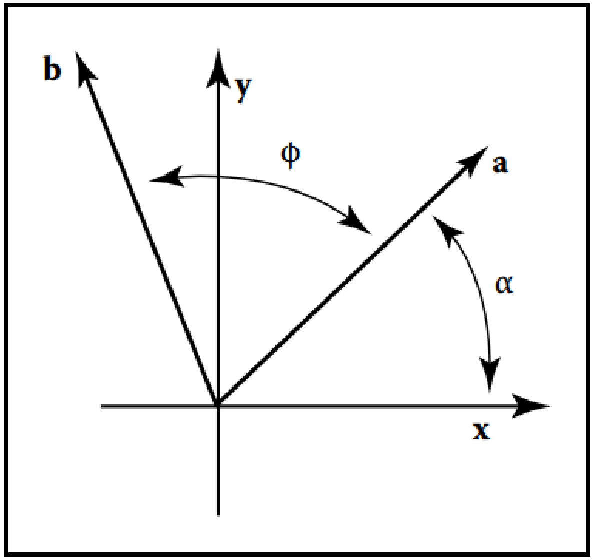
$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b = x_a\cos\phi - y_a\sin\phi,$$

$$y_b = y_a\cos\phi + x_a\sin\phi.$$

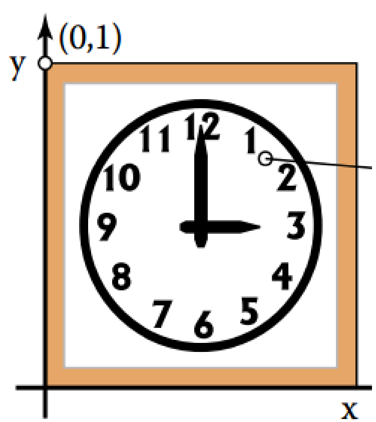
$$\text{rotate}(\phi) = \begin{bmatrix}\cos\phi & -\sin\phi \\ \sin\phi & \cos\phi\end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (7/11)

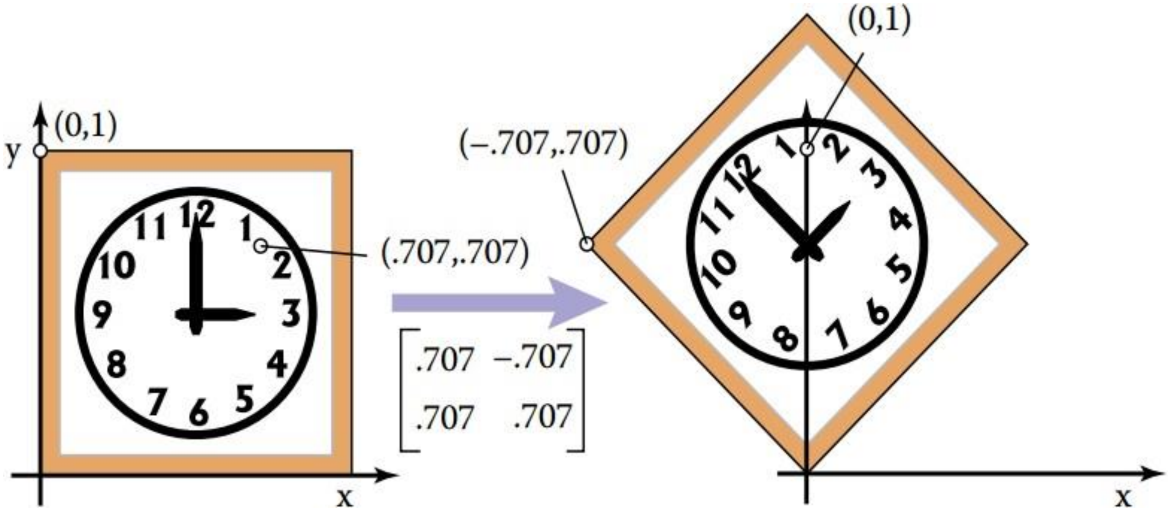
$$\begin{bmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (8/11)

$$\begin{bmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



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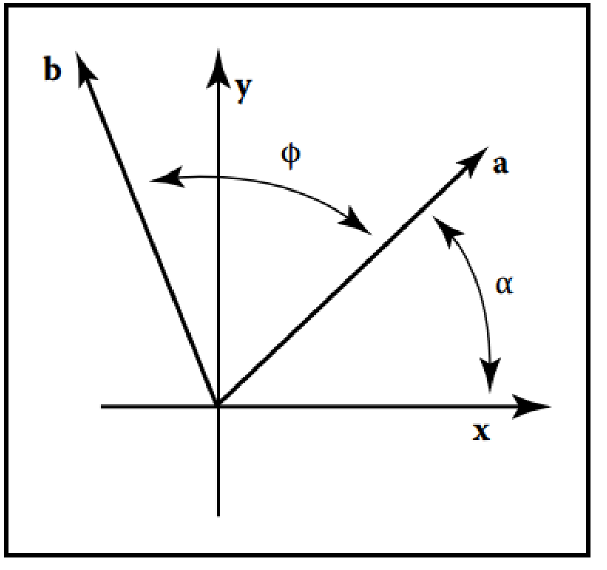
Rotation (9/11)

$$\text{rotate}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

Q: What about – ve angle?



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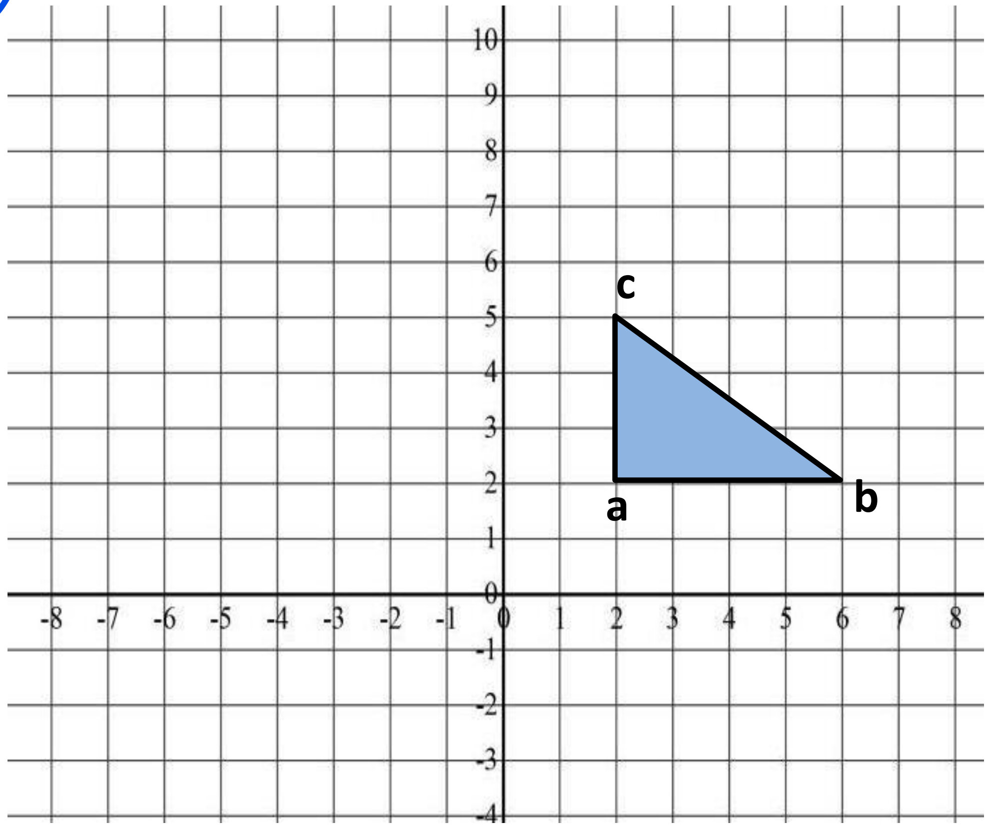
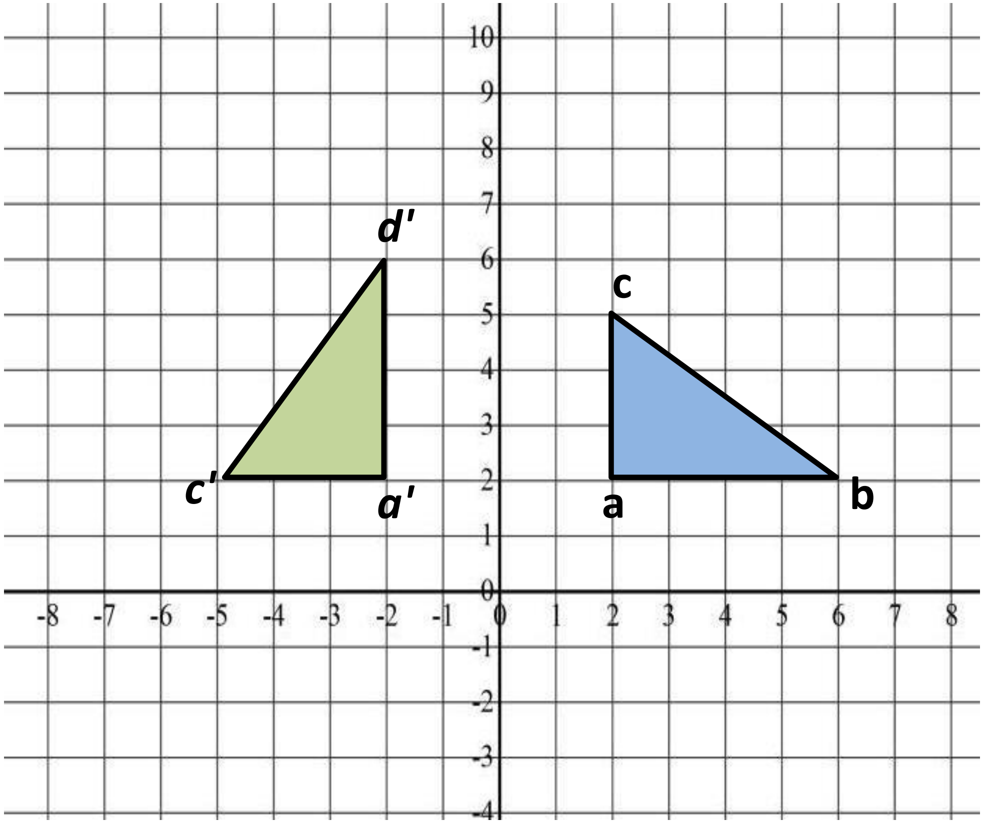
Rotation (10/11)

Rotate(90)



Rotation (11/11)

Rotate(90)

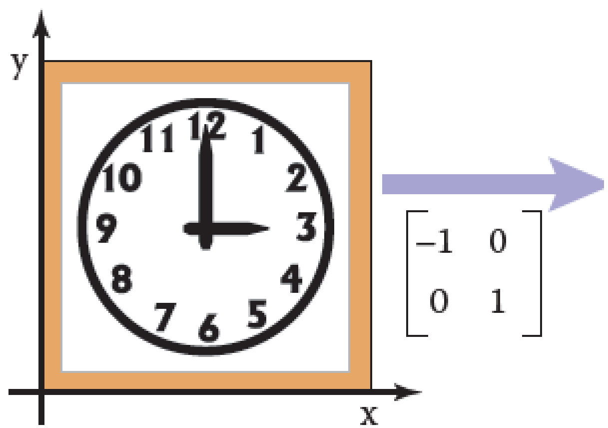


Reflection (1/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Reflection (2/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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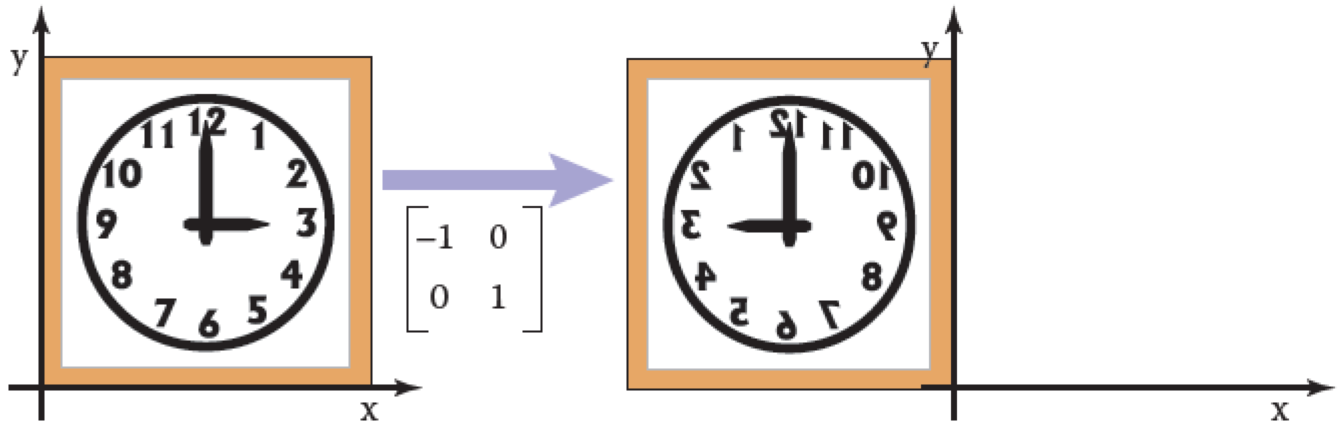
Reflection (3/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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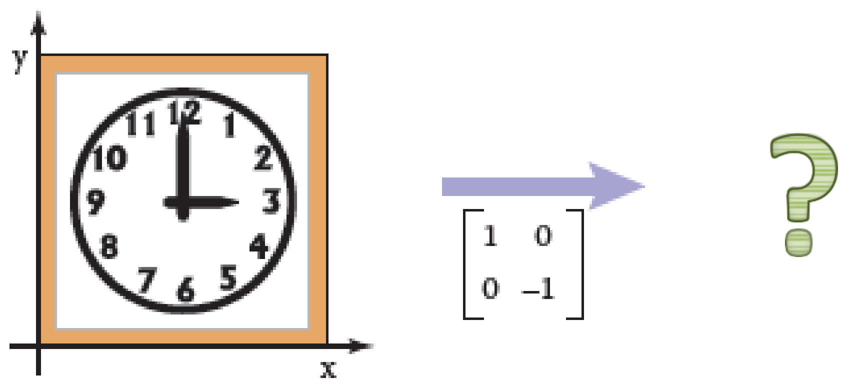
Reflection (5/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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Affine transformation (1/2)

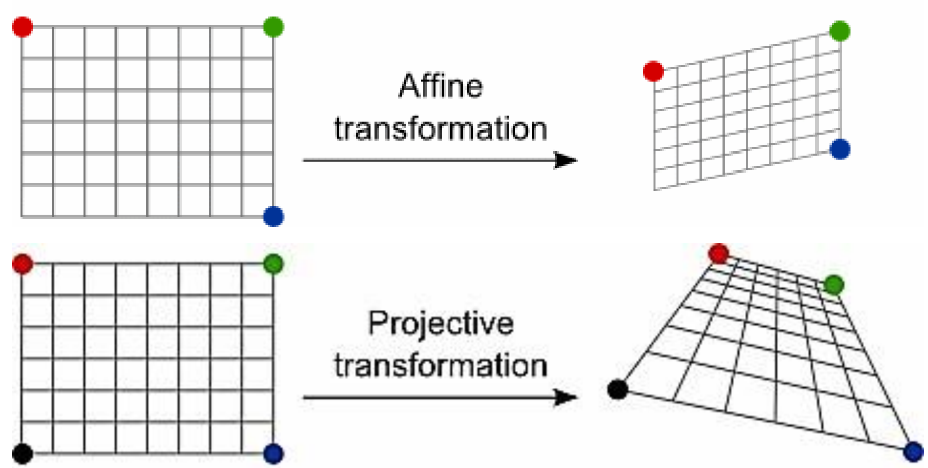
- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.

Example of Non-Affine Transformation:
A perspective transformation is a non-affine transformation. In a perspective transformation, parallel lines in the real world do not remain parallel after transformation. This is commonly seen in photography when objects appear smaller as they move farther away from the camera. Perspective transformations distort both angles and distances between points, making them non-affine.

Source: https://www.graphicsmill.com/docs/gm/affine-and-projective-transformations.htm | https://en.wikipedia.org/wiki/Affine_transformation

Affine transformation (2/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.



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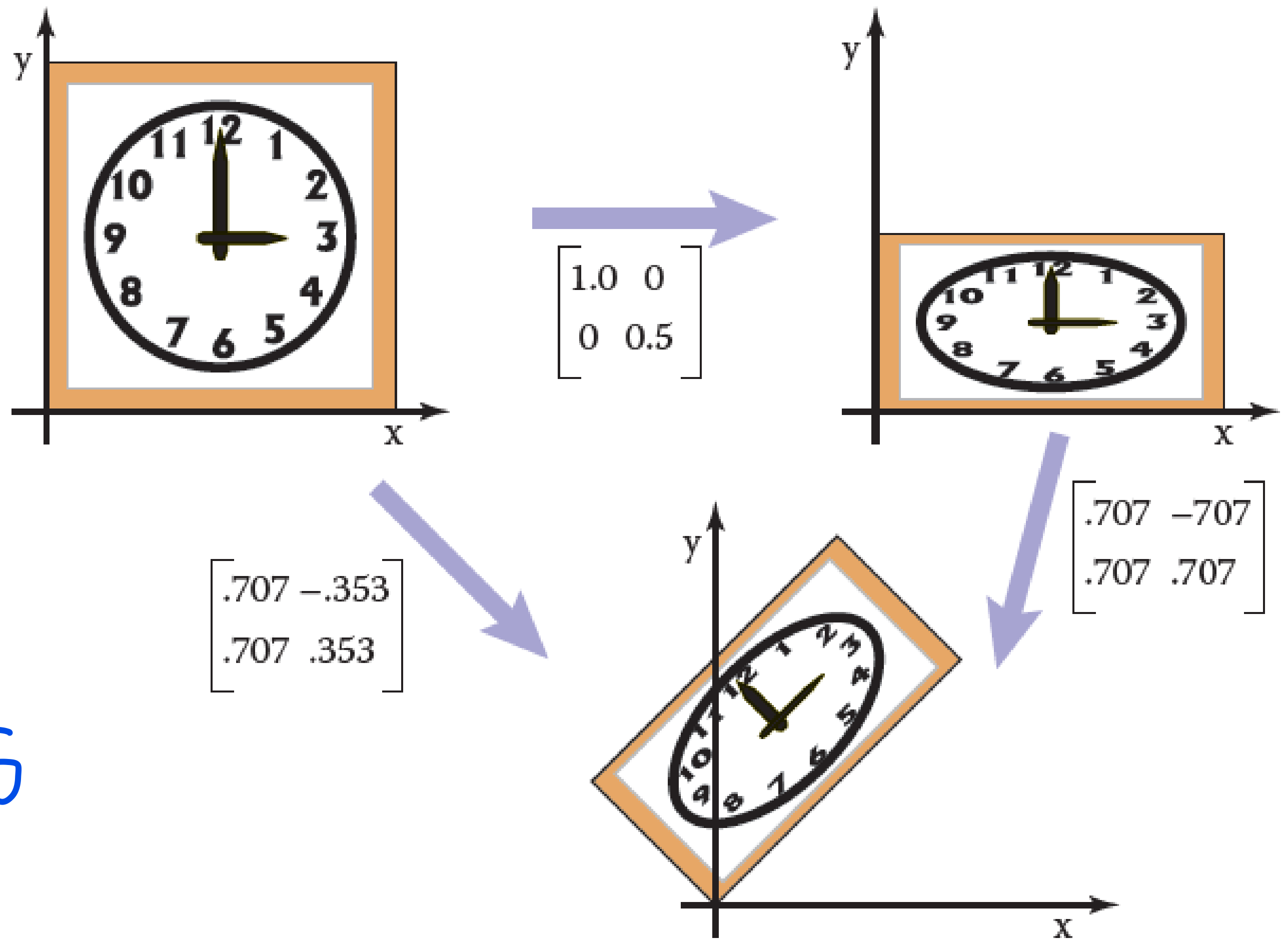
Composition of Transformations (1/11)



$M = R \times S$

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (2/11)

- Apply more than one transformation:
  – i.e., for a 2D point v_1 we might want to –
  1. first apply a scale S
  2. then a rotation R.
- This would be done in two steps:
  1. **first, $v_2 = S\,v_1$**
  2. **then, $v_3 = R\,v_2$.**

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (3/11)

Therefore –

1. $v_2 = S\, v_1$
2. $v_3 = R\, v_2$
3. $v_3 = R\,(Sv_1)$
4. $v_3 = (RS)\, v_1$ *[matrix multiplication is associative]*
5. $v_3 = Mv_1$ *[Where M=RS]*

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (4/11)

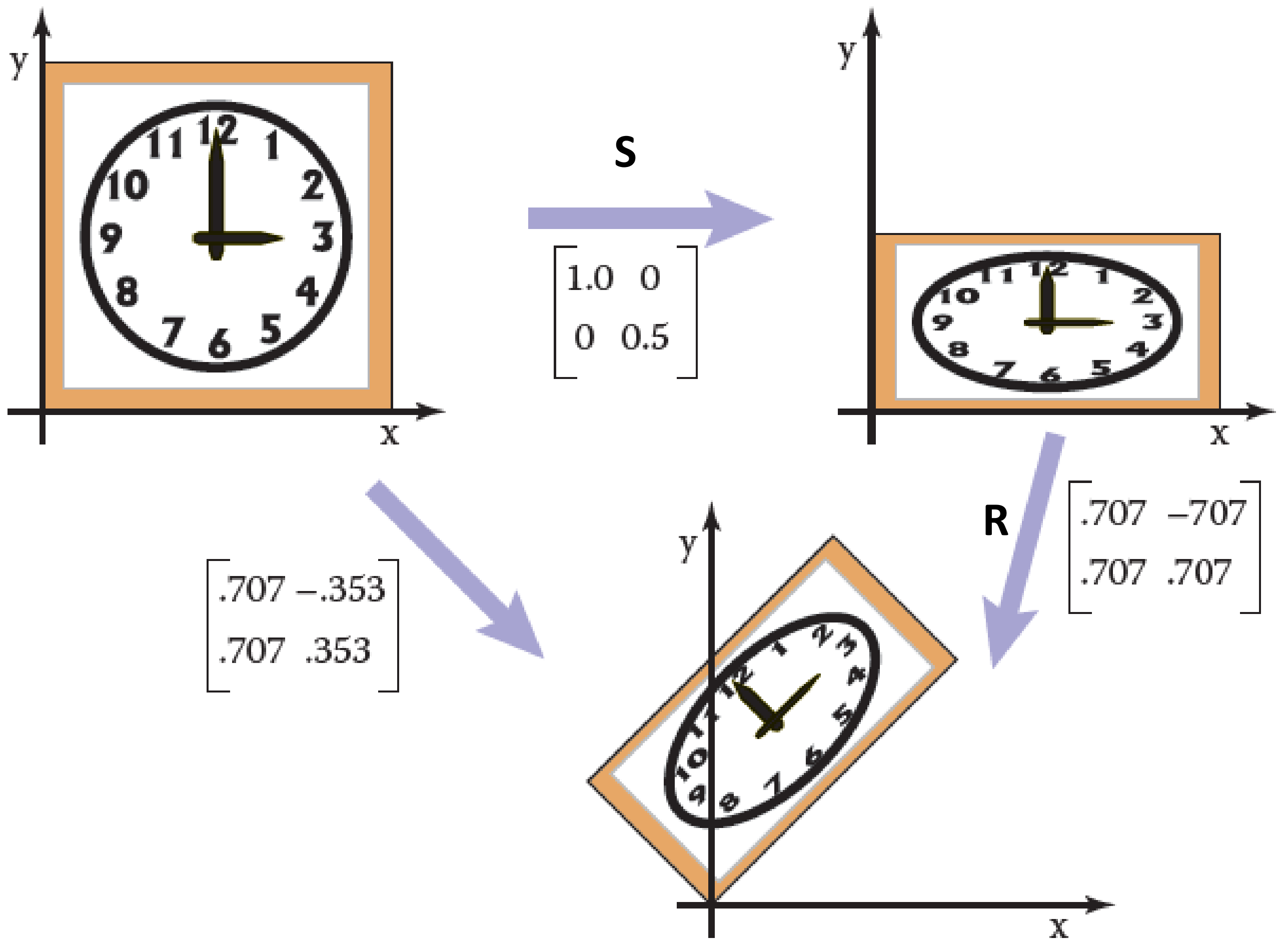
$$\mathbf{v}_{\mathbf{out}} = \mathbf{M}\,\mathbf{v}_{\mathbf{in}}$$

[Where M = R S]

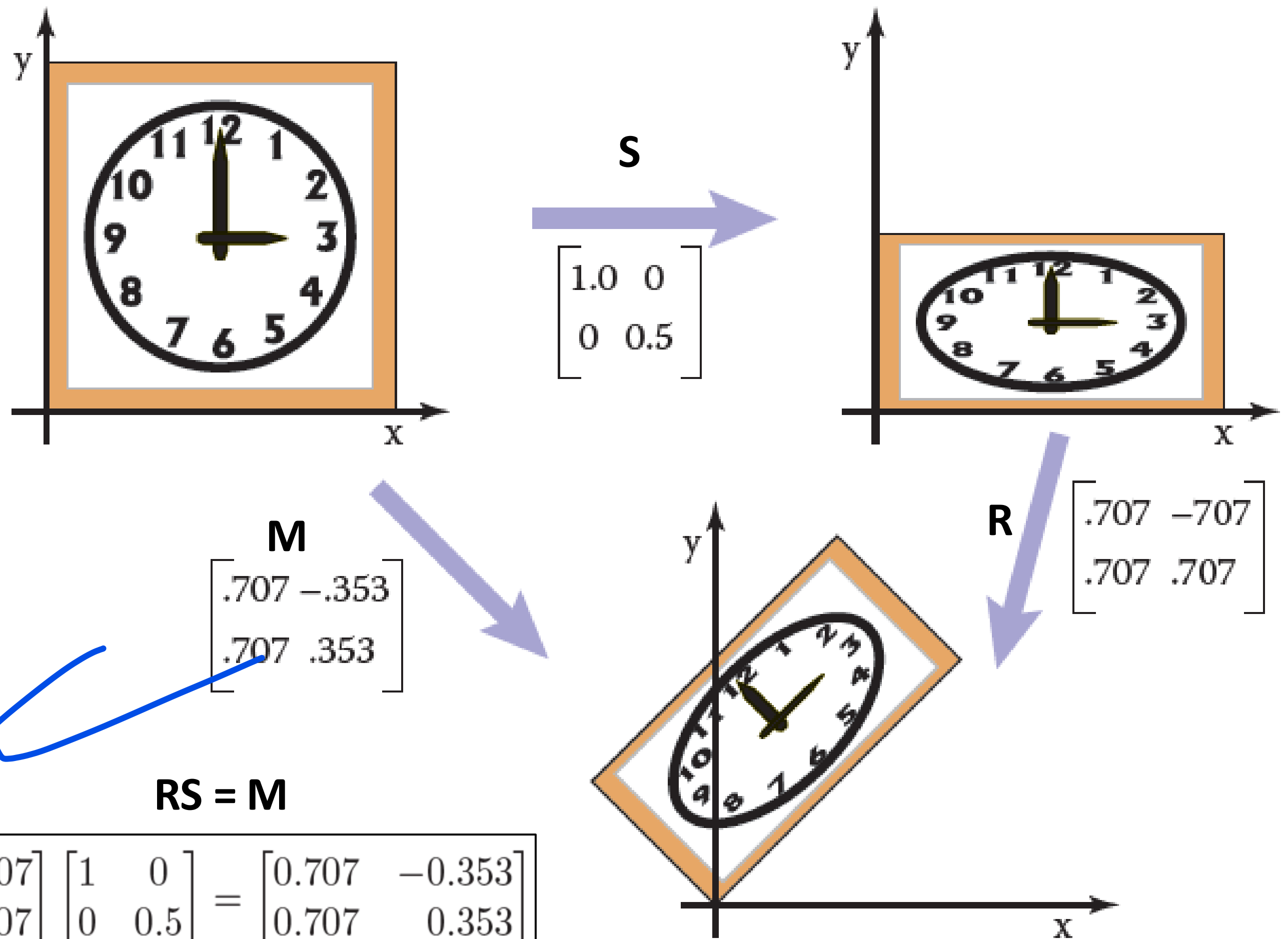
- We can represent the effects of transforming a vector by two matrices in sequence using a single matrix of the same size
  - which we can compute by multiplying the two matrices: **M = RS**

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (6/11)



Composition of Transformations (7/11)



RS = M

$$\begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.707 & -0.353 \\ 0.707 & 0.353 \end{bmatrix}$$

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

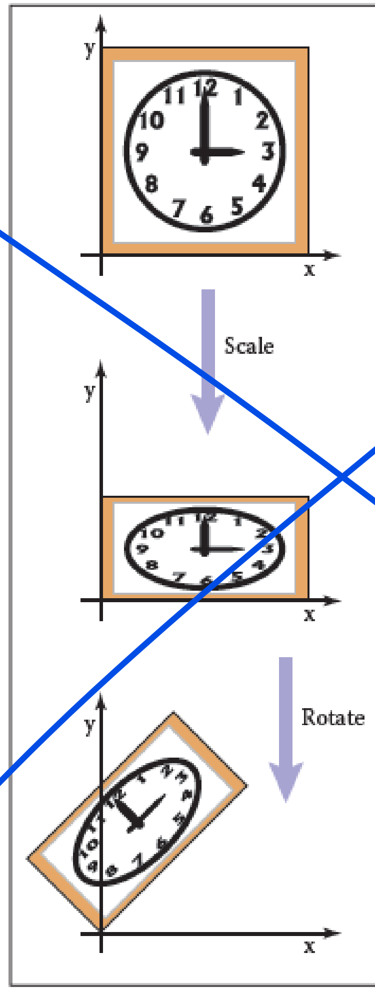
Composition of Transformations (8/11)

- It is very important to remember that these transforms are applied :
  - **from the right side first.**
  - So the matrix M= RS
    - first applies S and then R.

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (9/11)

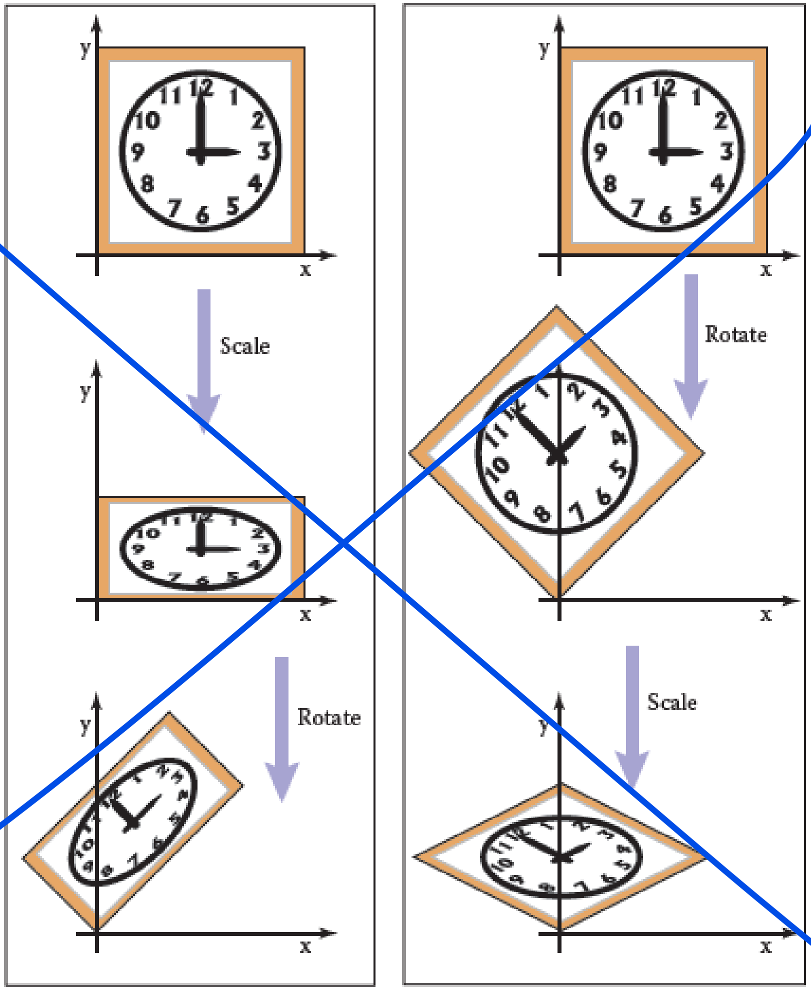
M= RS



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (10/11)

M= RS

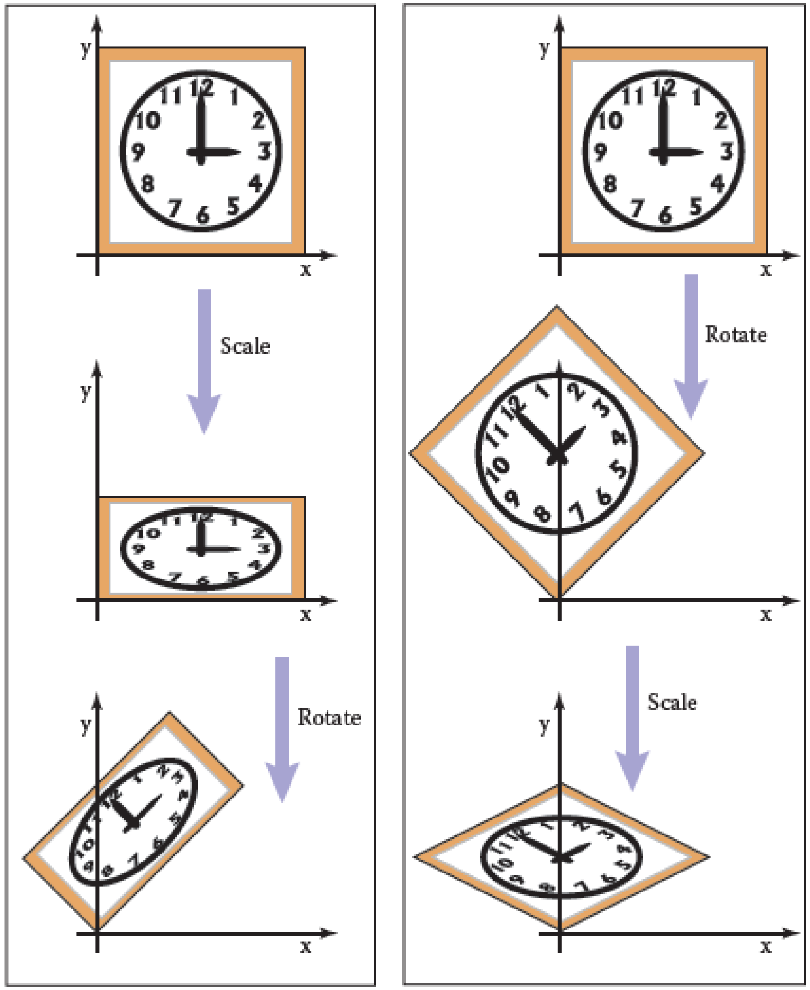


M = ?

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (11/11)

M= RS

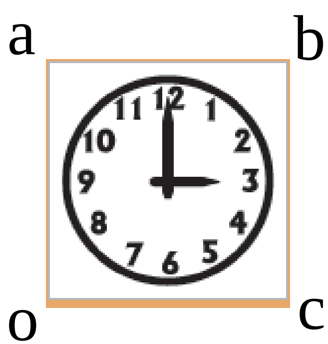


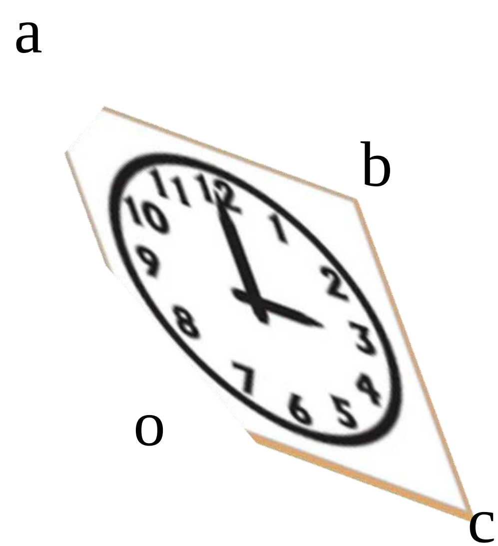
M= S R

Q: What about more than two transformations:
T1→T2→T3…. →Tn

Practice Problem - 1

- Stretch the clock by 50% along one of its diagonals
  – so that 8:00 through 1:00 move to the northwest and 2:00 through 7:00 move to the southeast.





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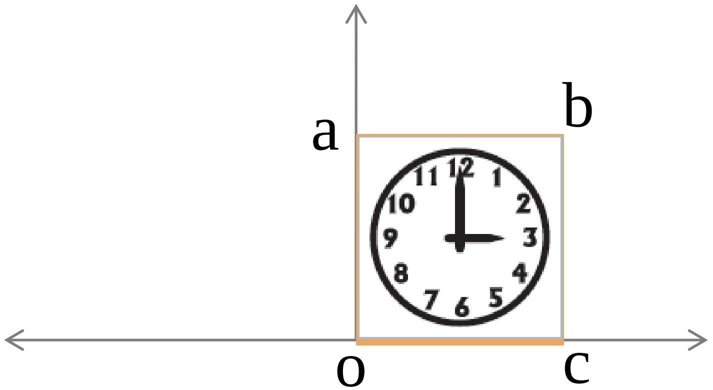
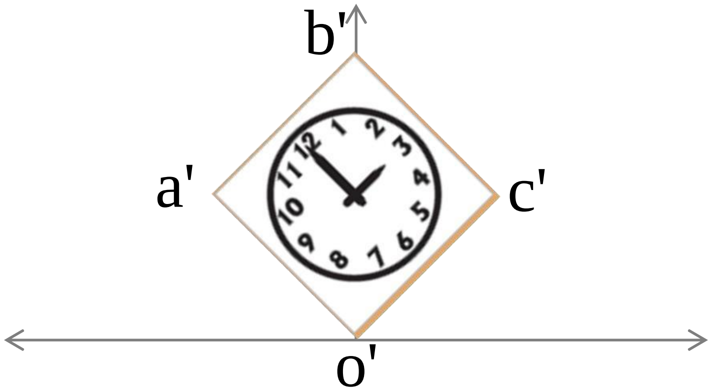
Practice Problem – 1 (Sol.)

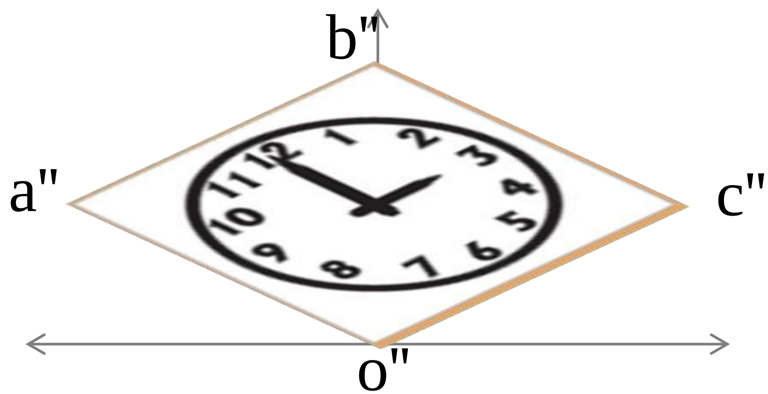


Practice Problem – 1 (Sol.)



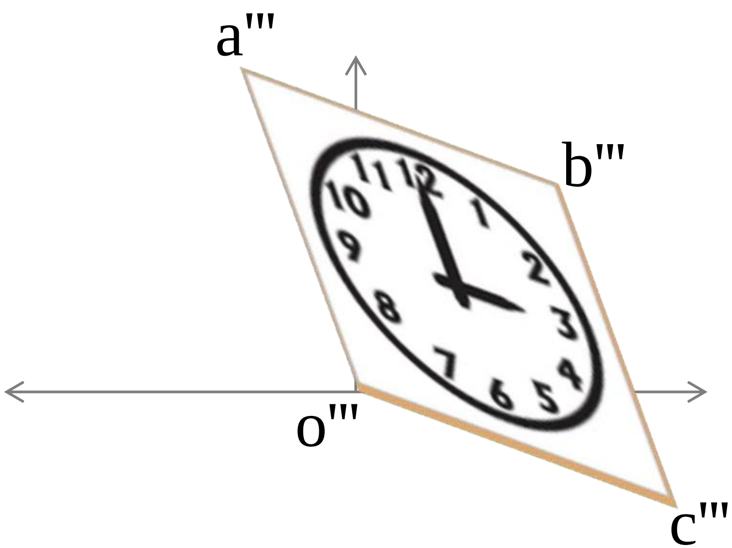
rotate(?)

Practice Problem – 1 (Sol.)



scale(?)

Practice Problem – 1 (Sol.)



rotate(?)

Practice Problem – 1 (Sol.)

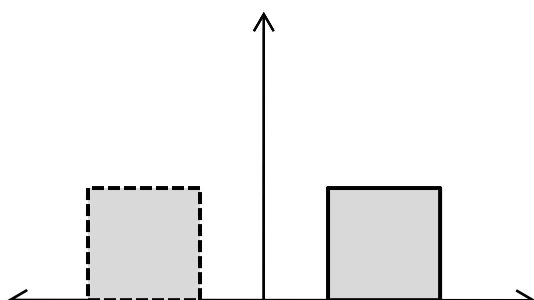
- rotate(45°) → scale(1.5, 1) → rotate(– 45°).
  - *Q: Draw the steps*
- $M = R(-45°)\ S(1.5, 1)\ R(45°)$

  $= R^T\ S\ R$
  - *Q: Calculate the matrix*

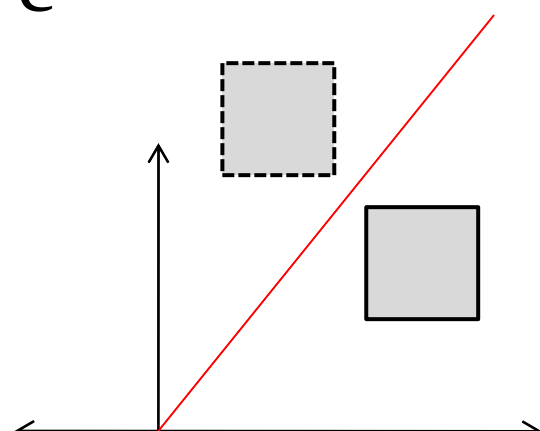
Practice Problem – 2

- Reflect the clock along a line goes through origin:

$$y = mx + c$$



w.r.t y-axis



w.r.t arbitrary line

Thank you

CSE4203: Computer Graphics

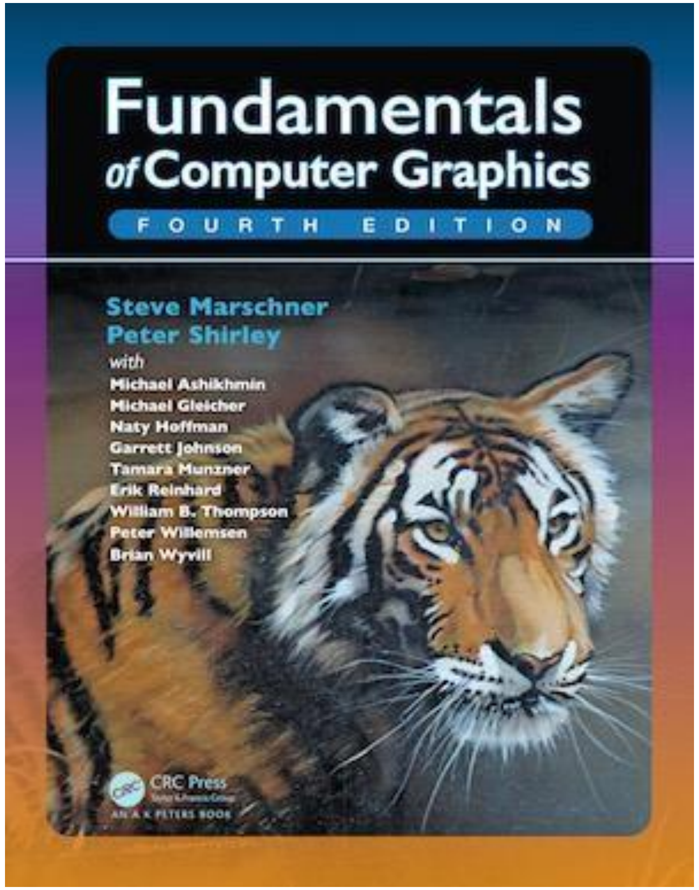
Chapter – 6 (part - B)

Transformation Matrices

Outline

– 3D Linear Transformation
– 3D Scaling
– 3D Rotation
– Translation
– Affine Transformation

Credit



CS4620: Introduction to Computer Graphics

Cornell University

Instructor: Steve Marschner

http://www.cs.cornell.edu/courses/cs4620/2019fa/

3D Transformation (1/1)

- The linear 3D transforms are an extension of the 2D transforms.
  - For 2D:

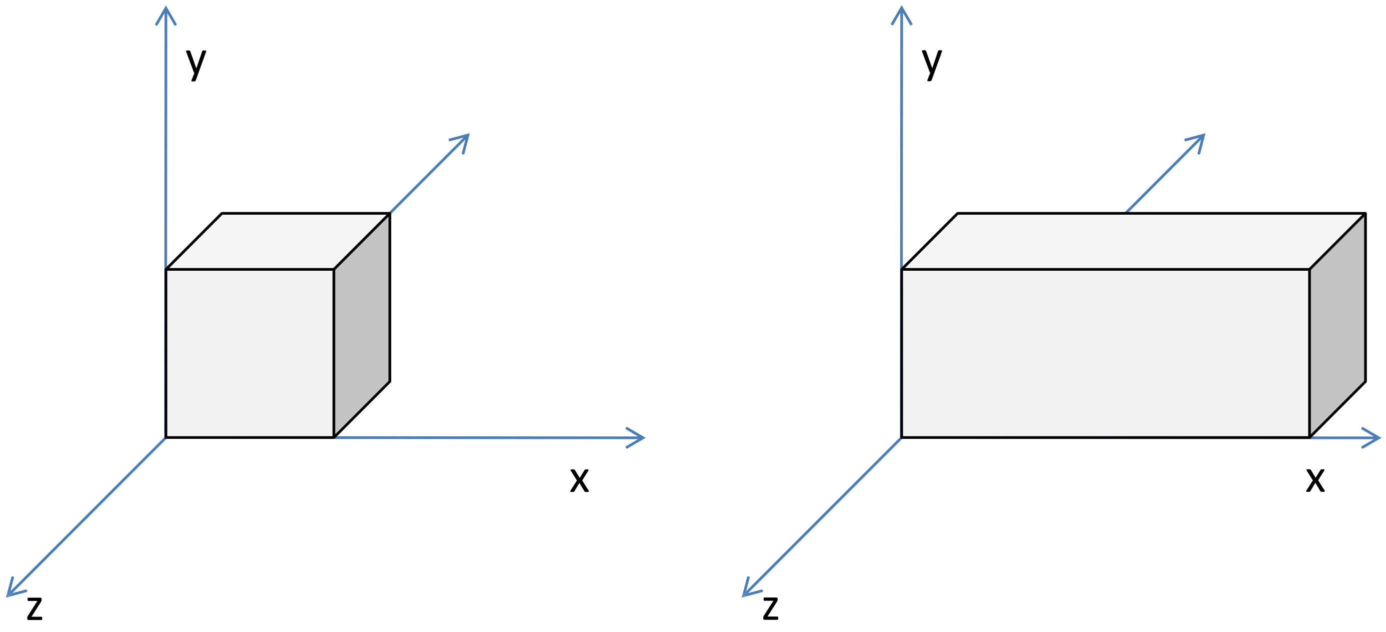
$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

  - For 3D:

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

3D Scaling (1/1)

$$\text{scale}(s_x, s_y, s_z) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}$$



3D Rotation (1/5)

- Rotation around axis
  - Counter-clockwise, w.r.t rotation axis.

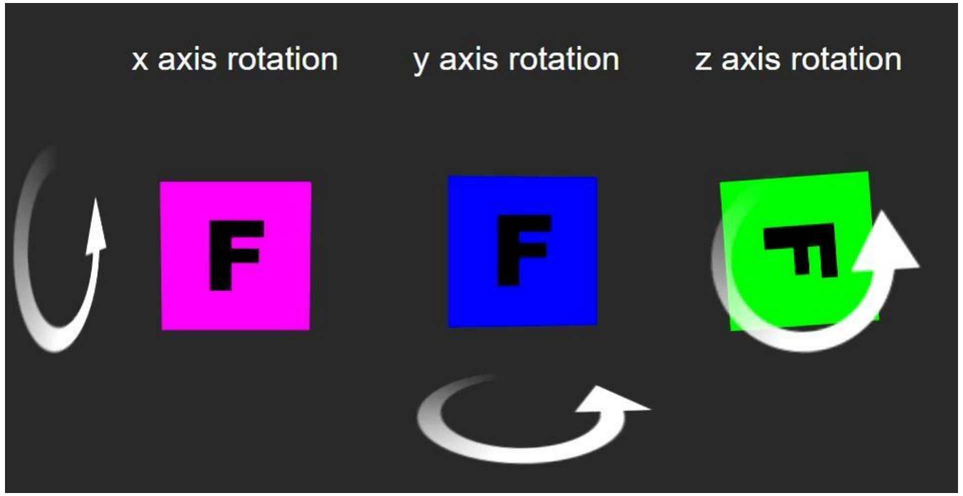
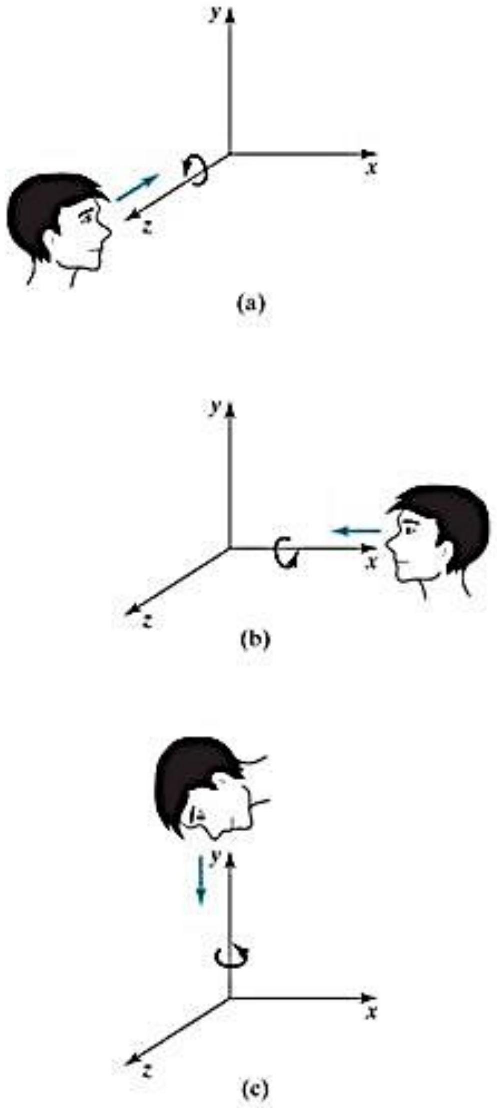
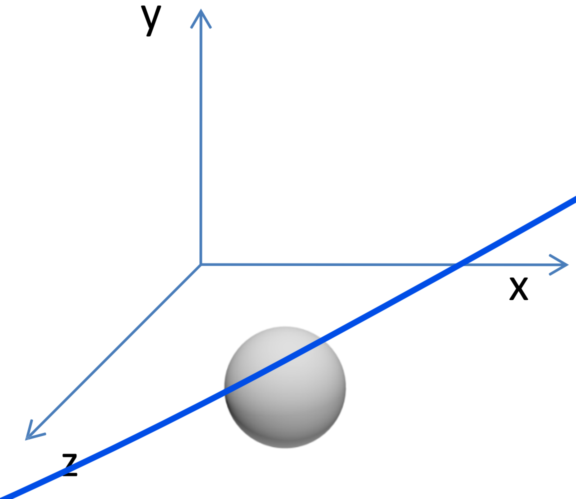


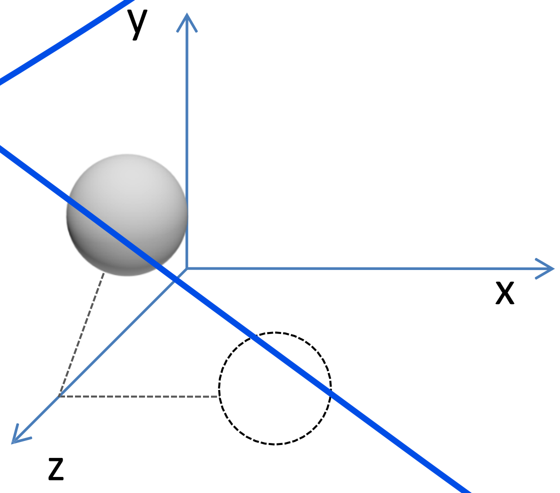
Image Source: https://slideplayer.com/slide/4889962/



3D Rotation (2/5)

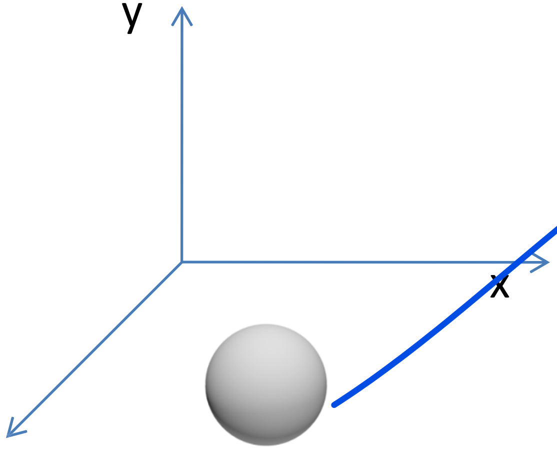
$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

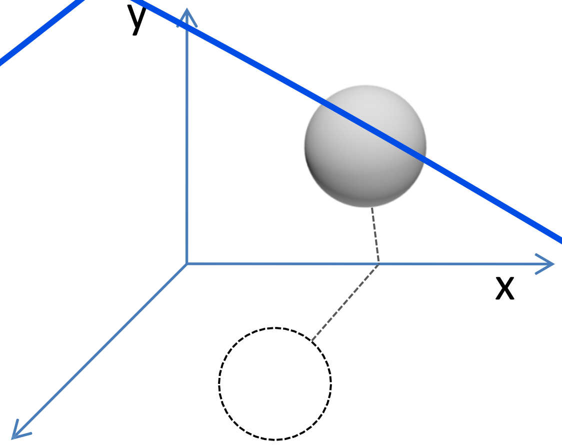




3D Rotation (3/5)

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$





3D Rotation (4/5)

$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

$$\text{rotate-y}(\phi) = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}$$

3D Rotation (5/5)

$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

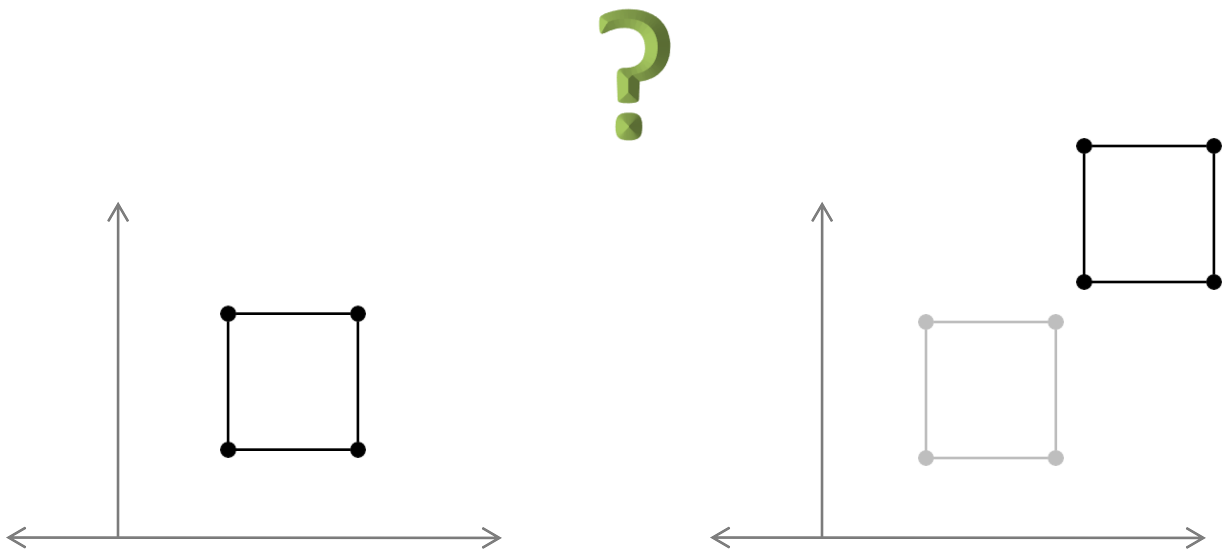
$$\text{rotate-y}(\phi) = \begin{bmatrix} \cos\phi & 0 & \boxed{\sin\phi} \\ 0 & 1 & 0 \\ \boxed{-\sin\phi} & 0 & \cos\phi \end{bmatrix}$$

Q: Why is it different?*

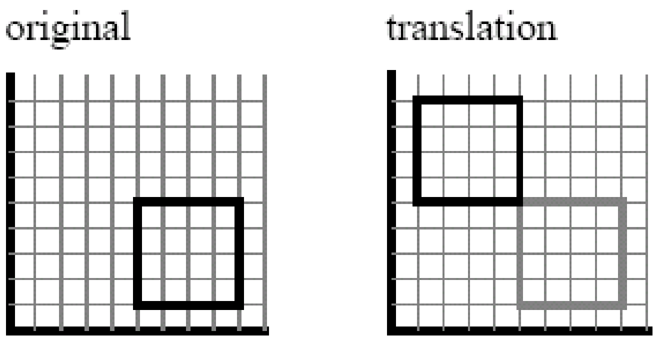
* https://robotics.stackexchange.com/questions/10702/rotation-matrix-sign-convention-confusion

Translation in 2D (1/8)

- Move or Translate to another position.



Translation in 2D (3/8)



$$\underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} + \begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{v'} = \mathbf{t} + \mathbf{v}}$$

$x' = x + t_x$
$y' = y + t_y$

$\mathbf{v'} = \mathbf{v} + \mathbf{t}$

Source: https://www.pling.org.uk/cs/cgv.html

Translation in 2D (4/8)

- But, for others cases, i.e. – scaling, rotation, we changed vectors **v** using a **matrix M**.
  - In 2D, these transforms have the form: -

$\begin{array}{rcccc} x' & = & m_{11}x & + & m_{12}y, \\ y' & = & m_{21}x & + & m_{22}y. \end{array}$	**v' = M v**

Translation in 2D (5/8)

- We **cannot** use such transforms to **translate**, only to scale and rotate them.

$$
\begin{aligned}
x' &= \cancel{m_{11}x} + m_{12}y, \\
y' &= \cancel{m_{21}x} + m_{22}y.
\end{aligned}
$$

$$\cancel{\mathbf{v' = M\,v}}$$

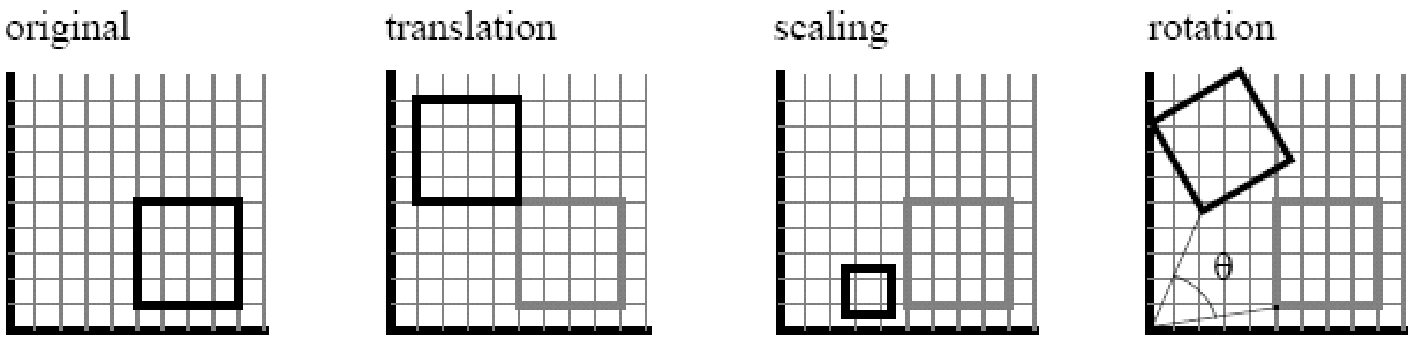
Translation in 2D (6/8)

- There is just no way to do that by **multiplying** (x, y) by a 2 × 2 matrix.
  - **adding** translation to our system of linear transformations:

$$\begin{aligned} x' &= m_{11}x + m_{12}y, \\ y' &= m_{21}x + m_{22}y. \end{aligned}$$	**v' = M v**
$$\begin{aligned} x' &= x + x_t, \\ y' &= y + y_t. \end{aligned}$$	**v' = v + t**

Translation in 2D (7/8)

- This is perfectly feasible –



$$\underbrace{\begin{bmatrix}x'\\ y'\end{bmatrix} = \begin{bmatrix}t_x\\ t_y\end{bmatrix} + \begin{bmatrix}x\\ y\end{bmatrix}}_{\mathbf{v' = v + t}} \qquad \underbrace{\begin{bmatrix}x'\\ y'\end{bmatrix} = \begin{bmatrix}s_x & 0\\ 0 & s_y\end{bmatrix} \cdot \begin{bmatrix}x\\ y\end{bmatrix} \qquad \begin{bmatrix}x'\\ y'\end{bmatrix} = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix} \cdot \begin{bmatrix}x\\ y\end{bmatrix}}_{\mathbf{v' = M\,v}}$$

Source: https://www.pling.org.uk/cs/cgv.html

Translation in 2D (8/8)

- This is perfectly feasible
  - But, the rule for **composing transformations** is not as simple and clean as with linear transformations.

$$T = T_n \,.\, T_{n-1} \ldots\ldots T_1 \,.\, T_0$$

$\begin{aligned} x' &= m_{11}x + m_{12}y, \\ y' &= m_{21}x + m_{22}y. \end{aligned}$	**v' = M v**
$\begin{aligned} x' &= x + x_t, \\ y' &= y + y_t. \end{aligned}$	**v' = v + t**

Homogeneous Coordinates (1/9)

– Instead, we can use a clever trick to get a single matrix multiplication to do both.

$$\begin{bmatrix} x' \\ y' \\ \end{bmatrix} = \begin{bmatrix} & & \\ & \text{2 x 2} & \\ & & \end{bmatrix} \begin{bmatrix} x \\ y \\ \end{bmatrix}$$

Homogeneous Coordinates (2/9)

- Instead, we can use a clever trick to get a single matrix multiplication to do both.

- The idea is simple: represent the point (x, y) by a 3D vector $[x\ y\ 1]^T$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} & & \\ & \text{3 x 3} & \\ & & \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (3/9)

  - Instead, we can use a clever trick to get a single matrix multiplication to do both.
- The idea is simple: represent the point (x, y) by a 3D vector **[x y 1]ᵀ**
- Use 3 × 3 matrices of the form.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (4/9)

- This kind of transformation is called an ***affine transformation***.
  - this way of implementing affine transformations by adding an extra dimension is called ***homogeneous coordinates***

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (5/9)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (6/9)

- Translation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_t \\ 0 & 1 & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (7/9)

- Scaling:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (8/9)

- Rotation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3D Transformation with Homogeneous Coordinates (1/1)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

2D/ 3D Transformations (1/3)

	2D	3D
T	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_t \\ 0 & 1 & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$
S	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} p & 0 & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$
R	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$RotX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta x) & -sin(\theta x) & 0 \\ 0 & sin(\theta x) & \cos(\theta x) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotY = \begin{bmatrix} \cos(\theta y) & 0 & sin(\theta y) & 0 \\ 0 & 1 & 0 & 0 \\ -sin(\theta y) & 0 & \cos(\theta y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotZ = \begin{bmatrix} \cos(\theta z) & -sin(\theta z) & 0 & 0 \\ sin(\theta z) & \cos(\theta z) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Inverse Transformations (1/2)

T	T^{-1}
$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -p \\ 0 & 1 & 0 & -q \\ 0 & 0 & 1 & -r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$
$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} p & 0 & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1/p & 0 & 0 & 0 \\ 0 & 1/q & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$
$RotX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\Theta x) & -sin(\Theta x) & 0 \\ 0 & sin(\Theta x) & \cos(\Theta x) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotY = \begin{bmatrix} \cos(\Theta y) & 0 & sin(\Theta y) & 0 \\ 0 & 1 & 0 & 0 \\ -sin(\Theta y) & 0 & \cos(\Theta y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotZ = \begin{bmatrix} \cos(\Theta z) & -sin(\Theta z) & 0 & 0 \\ sin(\Theta z) & \cos(\Theta z) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	?

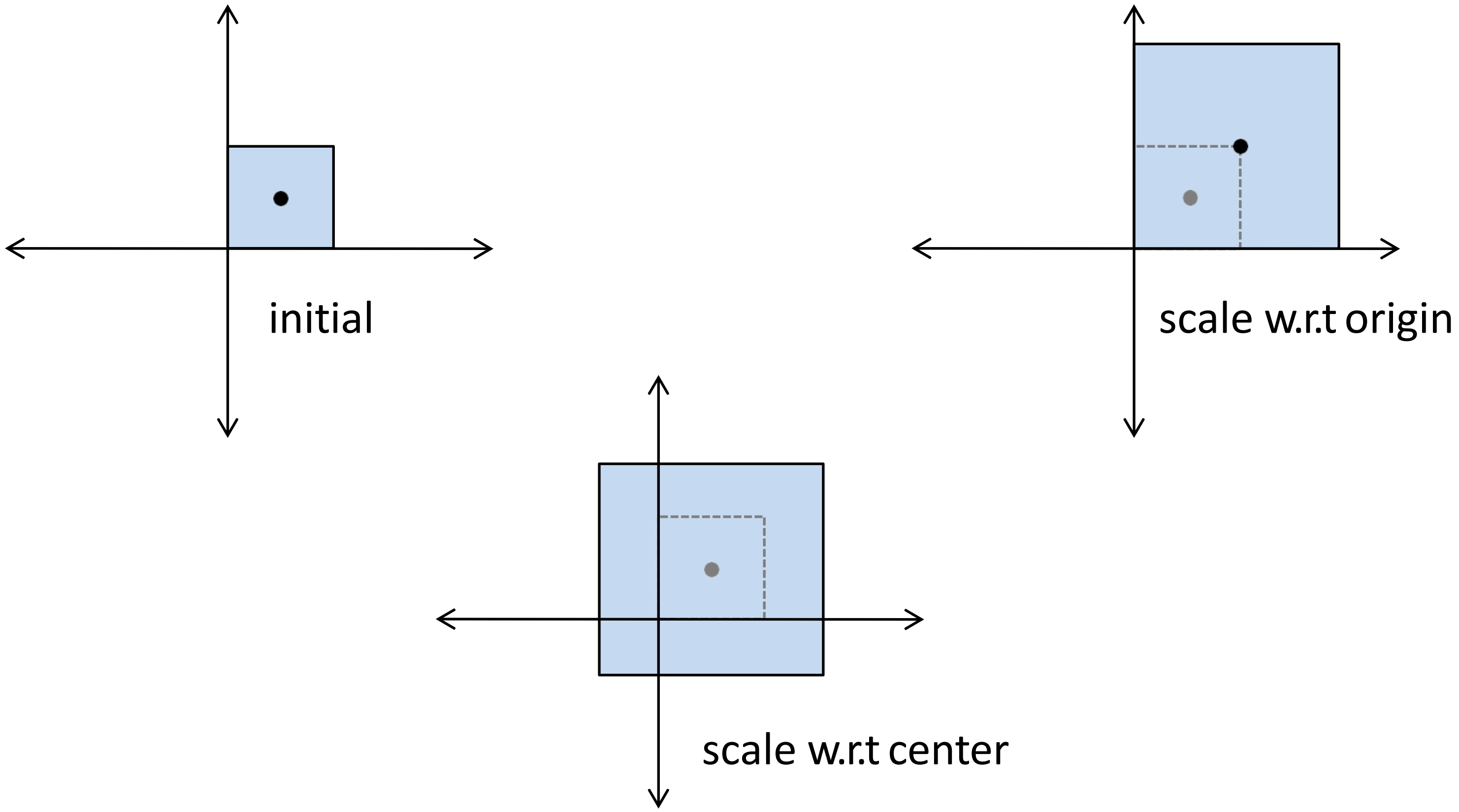
Inverse Transformations (2/2)

	Transformation	Inverse Transformation
T	T (tx, ty, tz)	$T^{-1} = T(-tx, -ty, -tz)$
S	S (sx, sy, sz)	$S^{-1} = S(1/sx, 1/sy, 1/sz)$
R	Rx(d) Ry(d) Rz(d)	$R^{-1} = R(-d) = R^T$ $Rx^{-1} = Rx^T$ $Ry^{-1} = Ry^T$ $Rz^{-1} = Rz^T$

Task: take any transformation matrix (i.e. scaling matrix ***S***) with numerical values, do the matrix inversion and see if it becomes S^{-1}

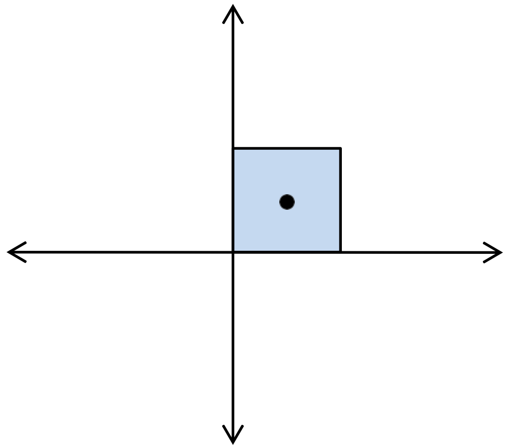
Practice Problem - 1

- Scale w.r.t the center



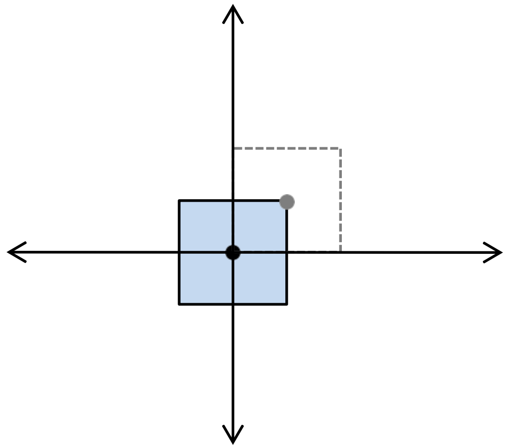
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



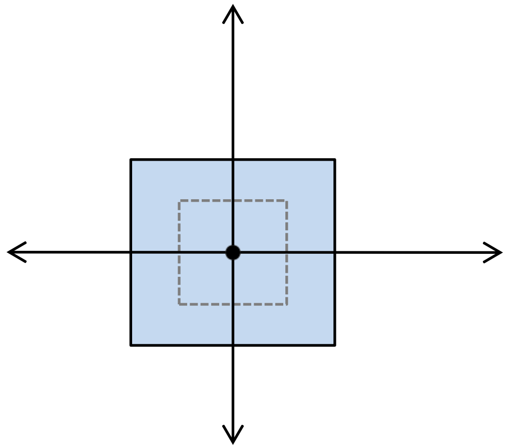
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



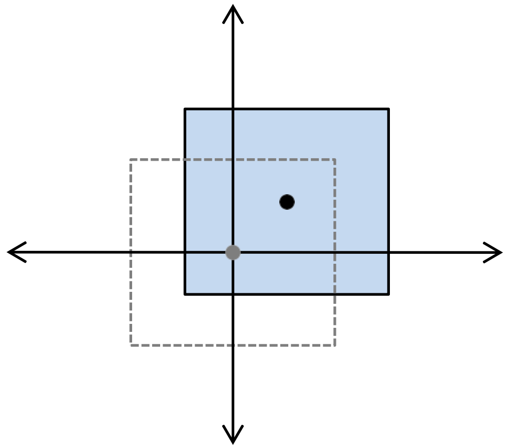
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



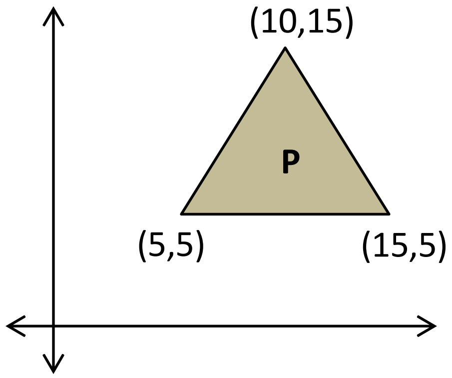
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



Practice Problem - 2

- We need to rotate a pyramid ***P*** about point **(5, 5)** by **90^o**. You have to –
  - Mention the steps to perform the task.
  - Determine the composite transformation matrix ***M***.
  - Multiply ***M*** with ***P*** and determine the new coordinates ***P'***.
  - Plot ***P*** and ***P'*** on the same axis to show the rotation.



Additional Reading

– 3D Shearing
– 3D reflection
– Rigid-body transforms
– Windowing transformations

Exercises

- Exercise 1 – 6, 8 and 9

CSE4203: Computer Graphics

Chapter – 6 (part - C)

Transformation Matrices

Outline

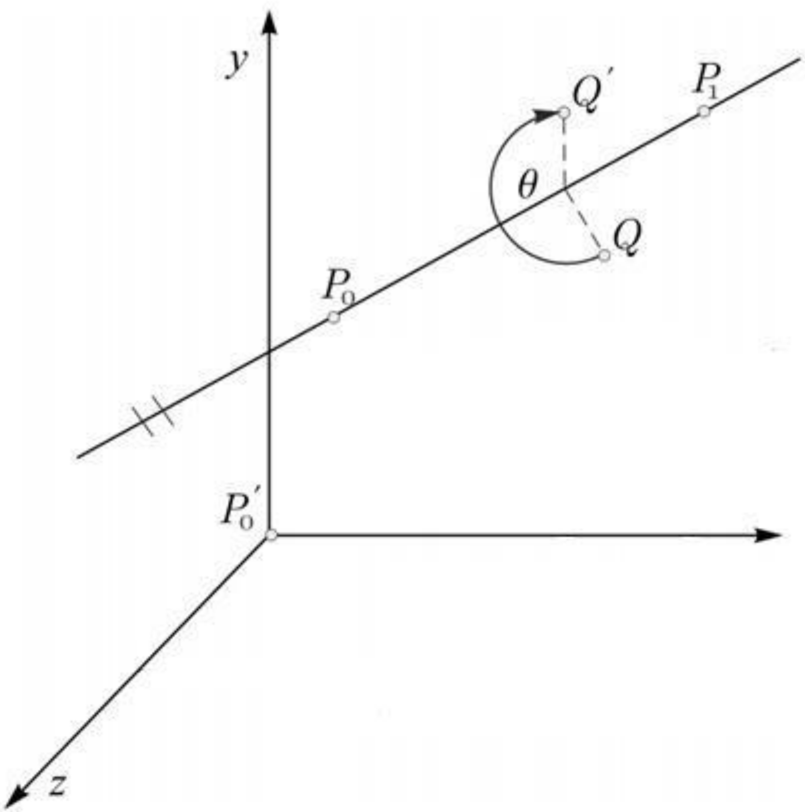
– 3D Transformation

– Rotation about an arbitrary line

References

- http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf
- http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Rotation about an arbitrary line (1/1)

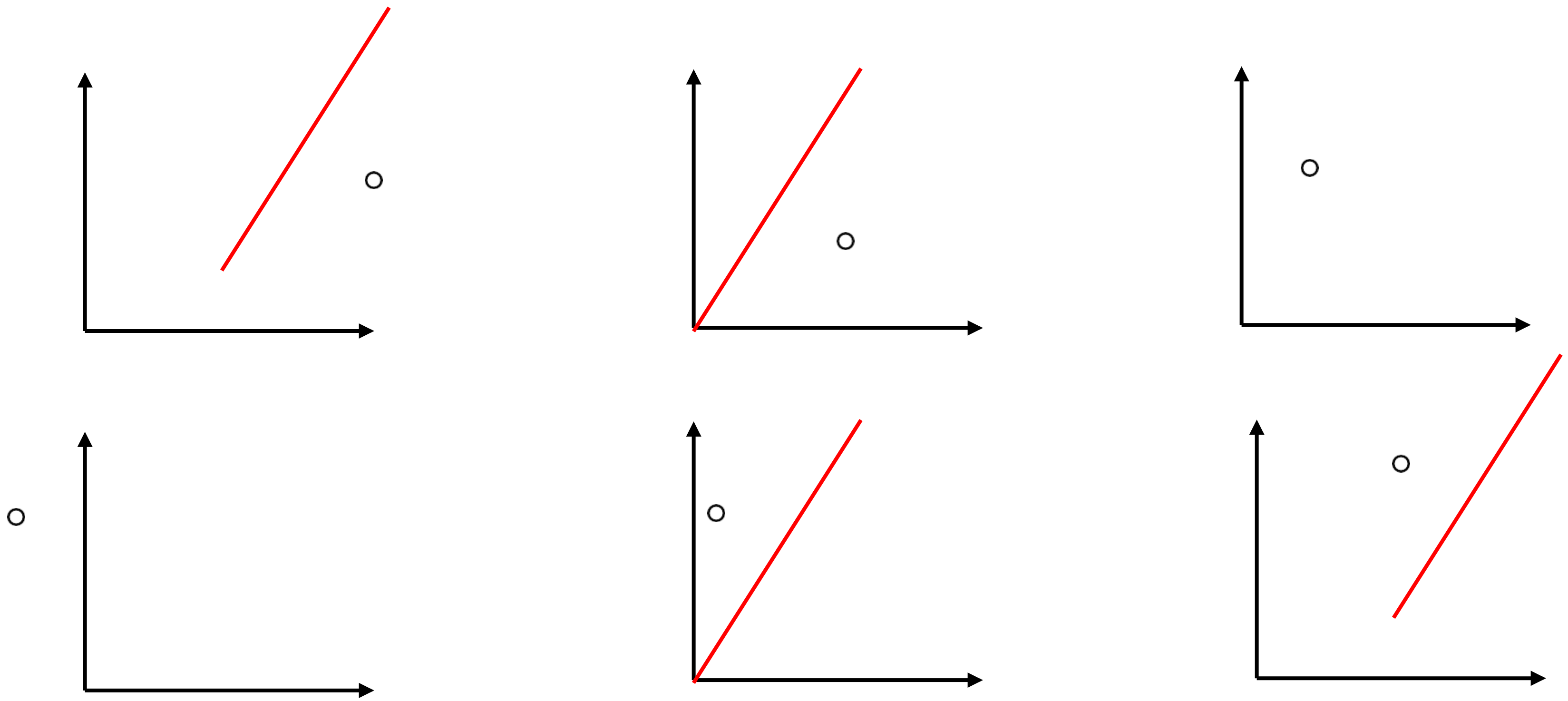


The basic idea is to make the arbitrary rotation axis coincide with one of the principle axis. Assume an arbitrary axis in space passing through the point **P0 (x0, y0, z0)** and **P1 (x1, y1, z1)** .

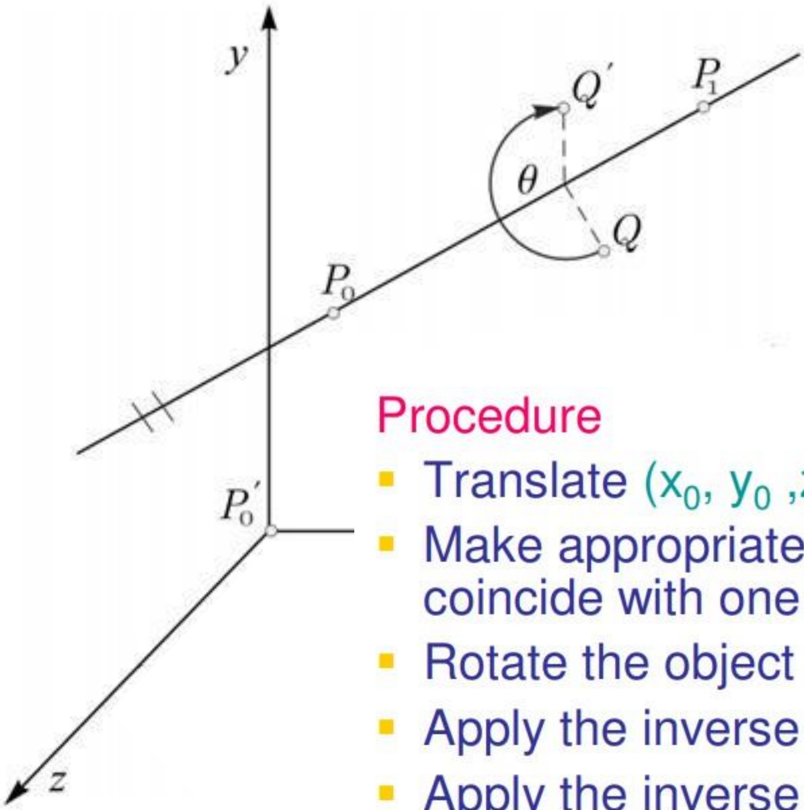
Credit: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf

In 2D case (1/1)

Reflecting about an arbitrary line



Steps (1/1)

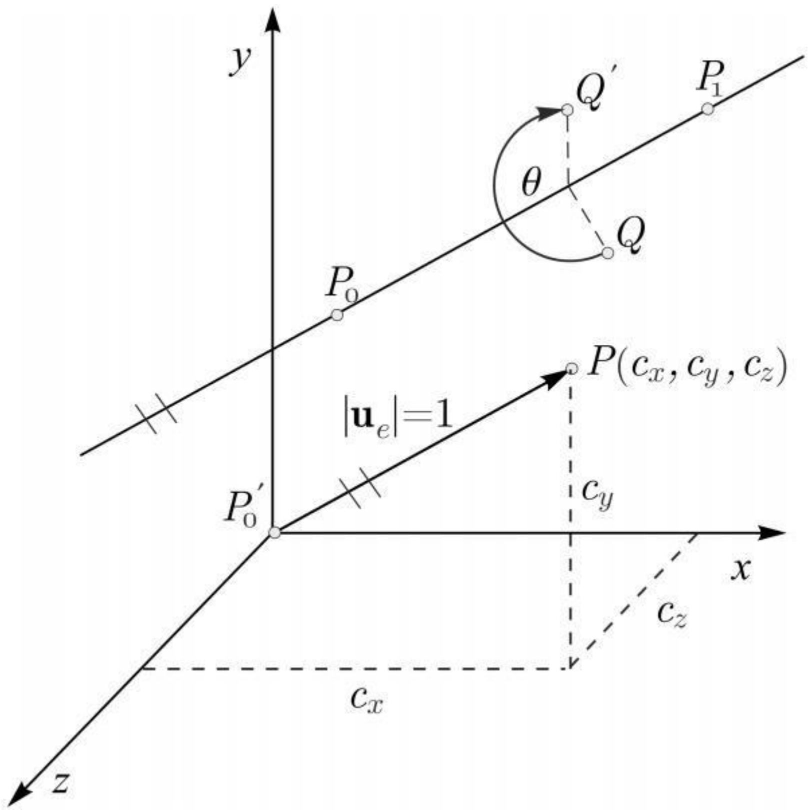


Procedure

- Translate (x_0, y_0, z_0) so that the point is at origin
- Make appropriate rotations to make the line coincide with one of the axes, say z-axis
- Rotate the object about z-axis by required angle
- Apply the inverse of step 2
- Apply the inverse of step 1

Credit: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf | http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Direction Cosine (1/2)

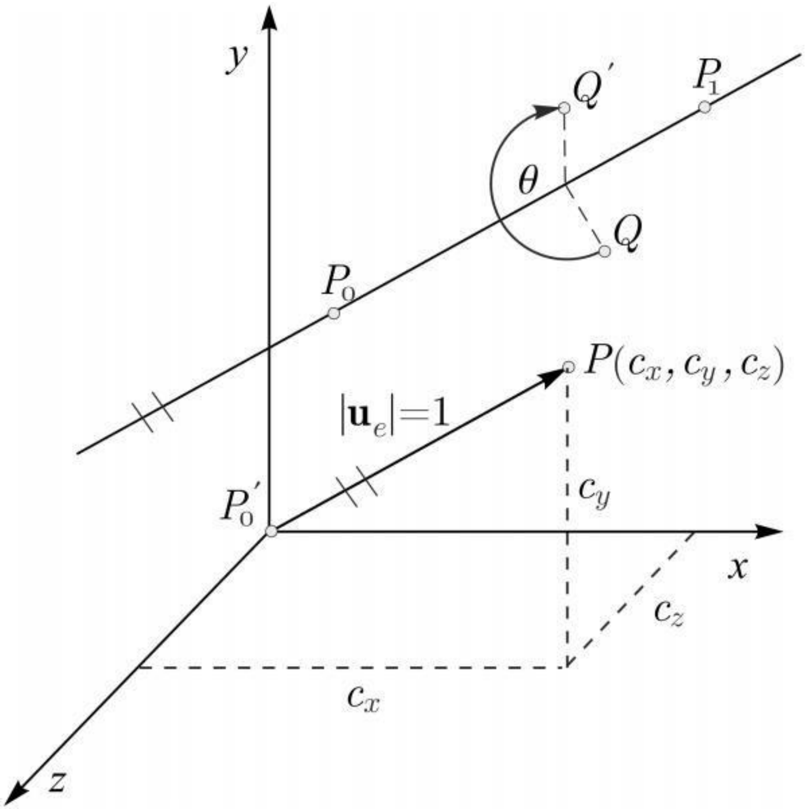


- Direction cosines of a vector are the cosines of the angles between the vector and the three coordinate axes.

Credit: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf

Direction Cosine (2/2)

Equivalently, they are the contributions of each component of the basis to a unit vector in that direction.



$$\mathbf{u} = P_1 - P_0$$

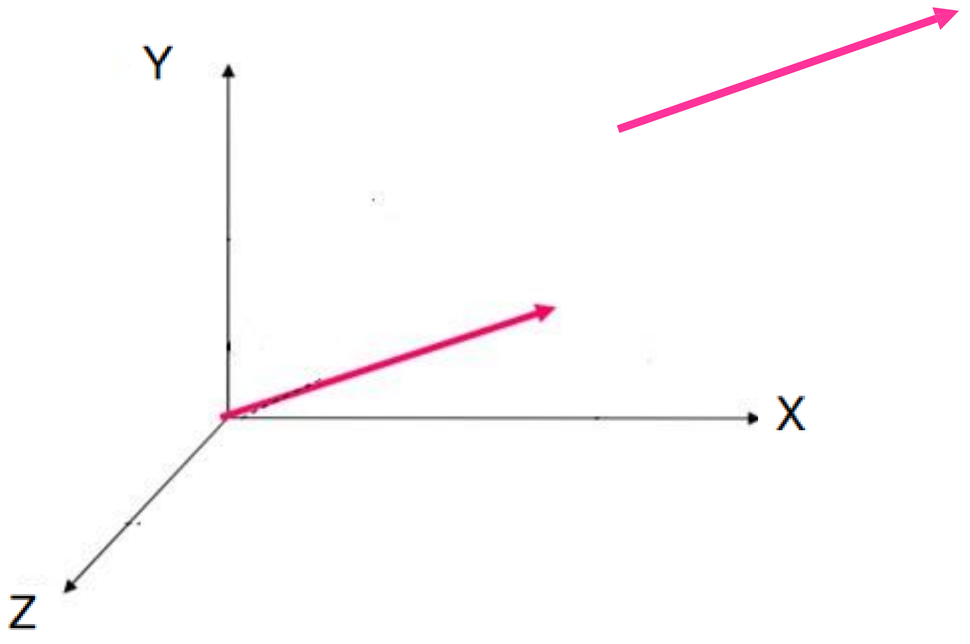
$$\mathbf{u_e} := \frac{\mathbf{u}}{|\mathbf{u}|} = (c_x, c_y, c_z)$$

$$c_x^2 + c_y^2 + c_z^2 = 1$$

$$\cos\phi_x = c_x, \quad \cos\phi_y = c_y, \quad \cos\phi_z = c_z.$$

Credit: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf

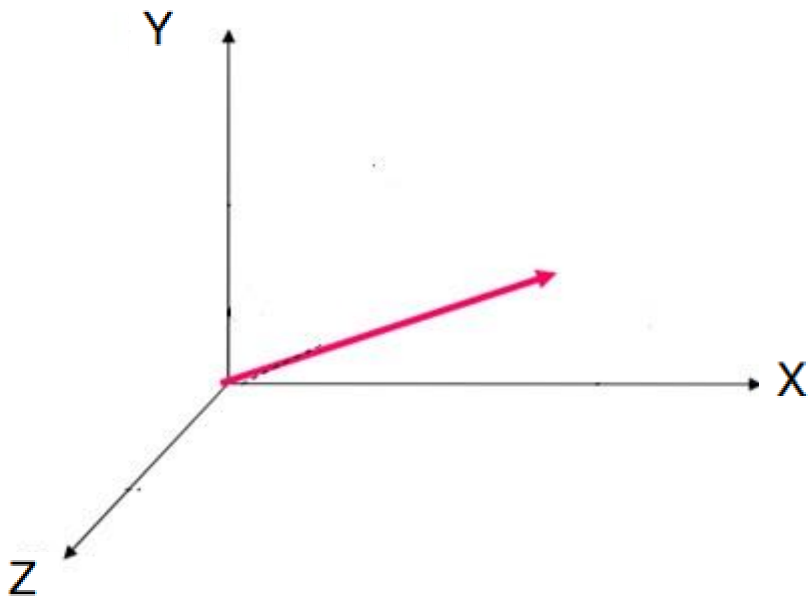
Coinciding the line with Principal axis (1/5)

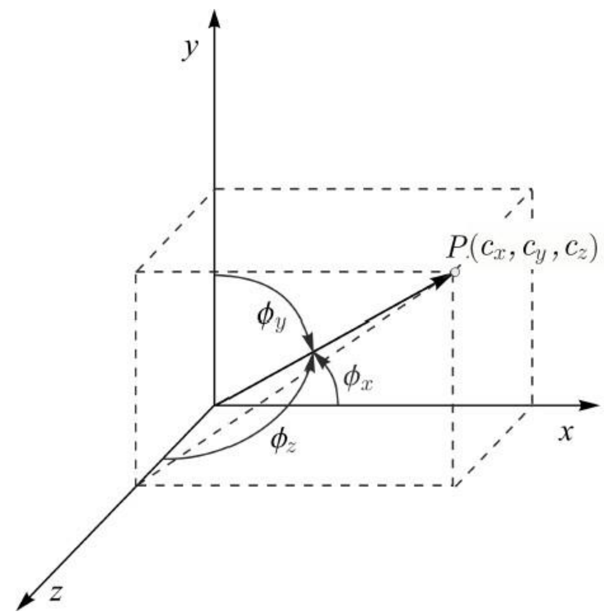


Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Coinciding the line with Principal axis (2/5)

Coinciding the arbitrary axis with any axis
the rotations are needed about other two axes

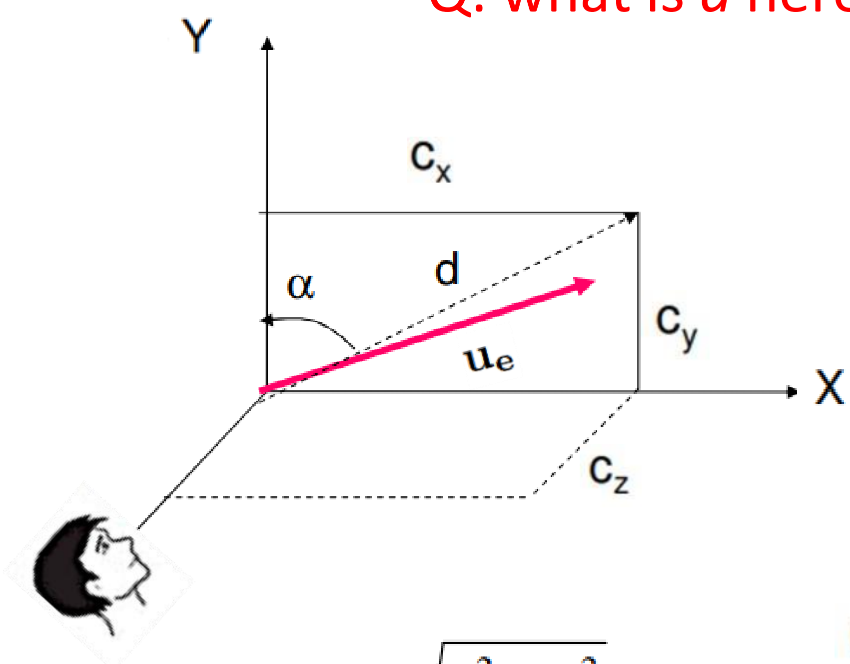


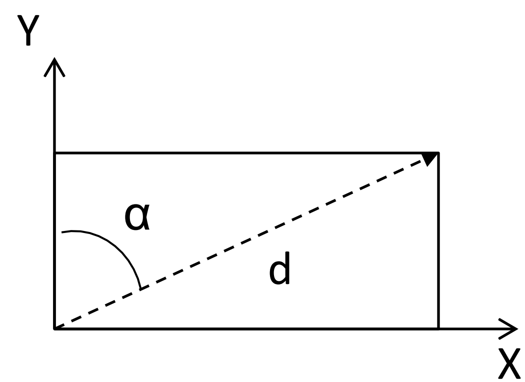


Source: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf | http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Coinciding the line with Principal axis (3/5)

Q: what is *d* here?

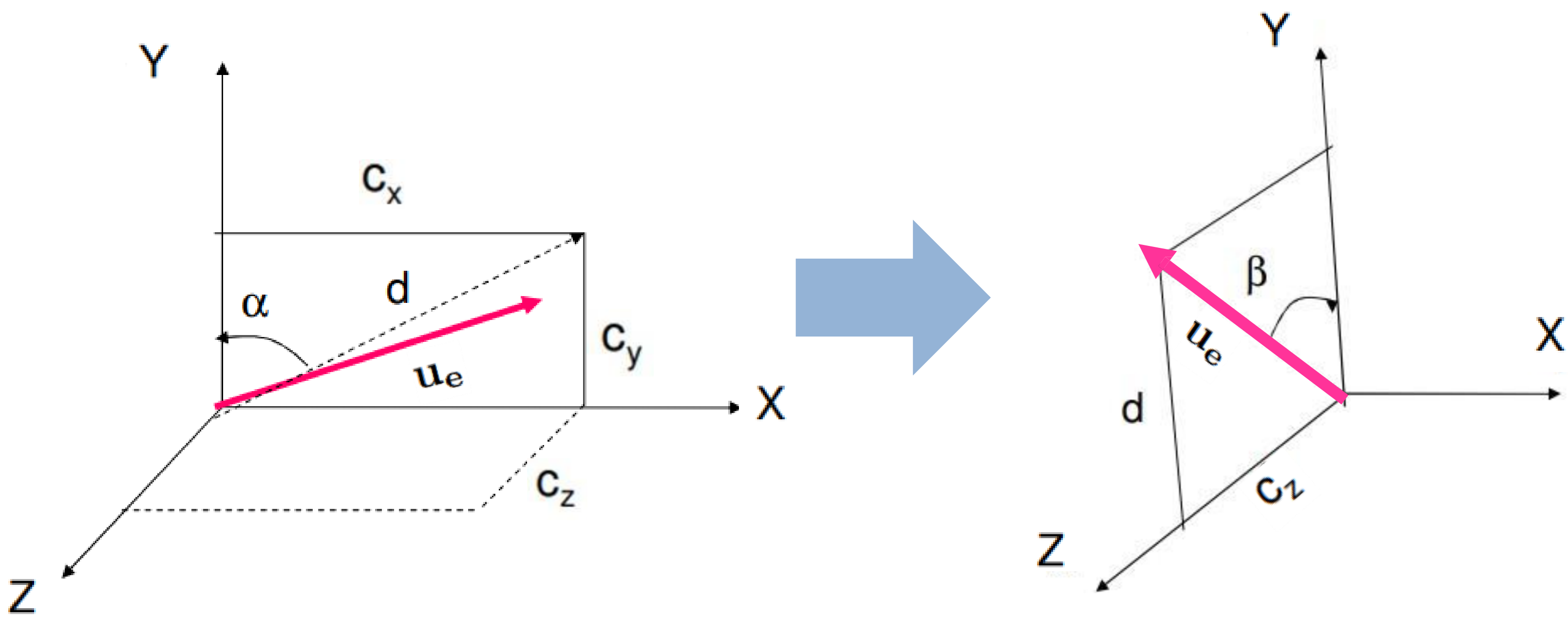




$$d = \sqrt{c_x^2 + c_y^2} \quad \cos\alpha = \frac{c_y}{d} \qquad \sin\alpha = \frac{?}{d}$$

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

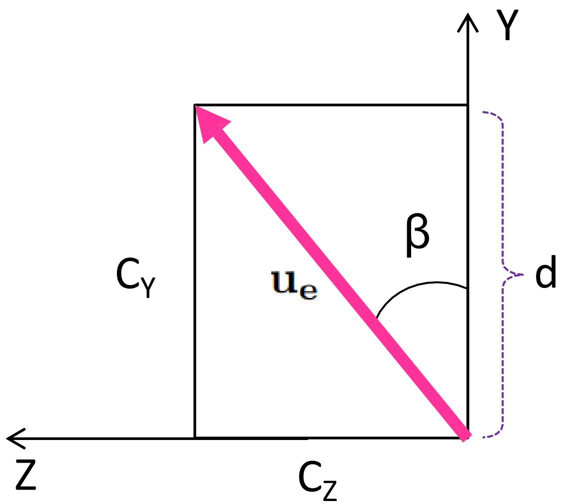
Coinciding the line with Principal axis (4/5)

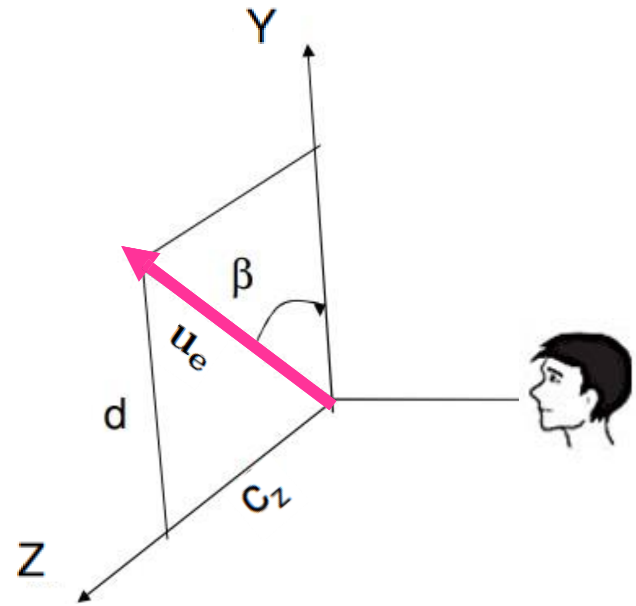


What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Coinciding the line with Principal axis (5/5)

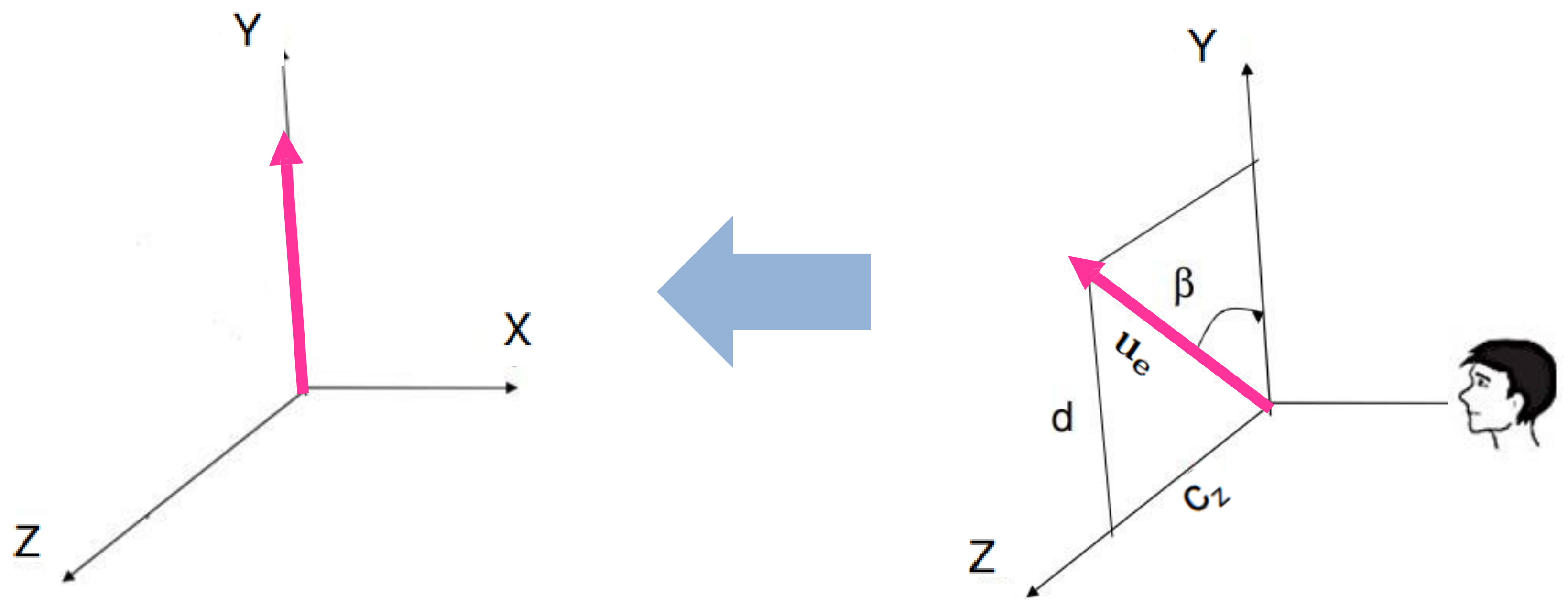




$\cos\beta = d$ $\sin\beta =$ **?**

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

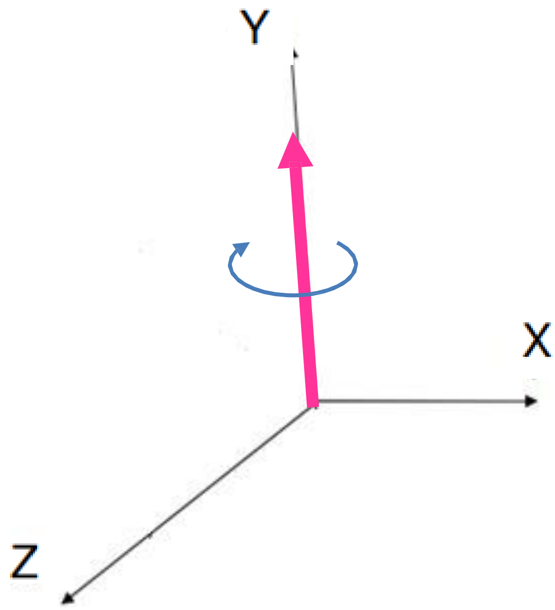
Rotating about the principal axis (1/2)



What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Rotating about the principal axis (2/2)



What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Undoing the steps (1/1)

- Q: What are the undoing steps?

Composite Transformation (1/1)

- **$M = T^{-1} * R_x^{-1}(-\beta) * \ldots ?$**

- **$Q' = M*Q$**

Practice Problem

- AB is a line and P is a point in 3D space; where the points A,B and P are (1,1,1), (3,3,4) and (2,2,4) respectively. We want to rotate P along AB by +90 degree. Determine the composite transformation matrix to do the task and calculate the rotated point P'.